Quantum Measurement, Minkowski Microstates and Matter Channels

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Abstract

Thermodynamic approaches to gravity, from Jacobson to Verlinde, leave unspecified the microscopic carriers of individual entropy increments. We show that the process of quantum measurement naturally models a gravitational microstate by developing an information-theoretic language of *handoff events*, *matter-memory*, and *spatial bits*, in which matter and horizon are complementary timelike and spacelike channels. Eliminating temperature between Landauer's principle and the Unruh effect yields an acceleration-dependent bound on the energy associated with information. A single construction, a helicity-anticorrelated Bell state in flat Minkowski space, makes contact with Jacobson's thermodynamic derivation of the Einstein equations, Van Raamsdonk's entanglement-generated geometry, the BMS supertranslation structure at horizons, a Carnot cycle, and effective field theory.

1 Introduction

Thermodynamic approaches to gravity are remarkably successful and remarkably incomplete. Jacobson derives the Einstein equations from entropy and temperature on local Rindler horizons [1]; Verlinde derives gravitational forces from entropy gradients on holographic screens [2]. Both frameworks operate at the level of bulk thermodynamics and neither specifies what physical process underlies each individual entropy increment. This paper proposes an answer: quantum measurement events, modeled as energy-momentum transfers under proper acceleration, provide explicit realizations of those increments in flat Minkowski space.

The physical reasoning is as follows: any measurement is mediated by a physical interaction, so information acquisition necessarily involves an exchange of energy and momentum; by the equivalence principle, this impulse is locally indistinguishable from acceleration in a gravitational field. We do not claim to have a complete quantum gravity program; instead we provide one explicit microstate construction, measurements of a helicity-anticorrelated Bell state sourced by a bilocal term in flat Minkowski space, and situate it in the current literature.

That a single simple construction — quantum measurement as energy–momentum transfer under proper acceleration, realised by one explicit photon pair in flat Minkowski space — makes simultaneous contact with three independent programs suggests that measurement events of this kind may describe the elementary process those frameworks collectively require. We do not claim this construction is unique or irreducible; we claim it is sufficient and concrete. Jacobson [1] derives the Einstein equations from thermodynamic

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$$
\begin{tikzcd}
\text{Local Gravitational Effects} \arrow[r, dashed, "\text{Diósi–Penrose}"] & \text{Quantum State Reduction} \\
\text{Acceleration} \arrow[u, "\text{local equivalence}"'] \arrow[r, "\text{energy–momentum transfer}"'] & \text{Observation} \arrow[u, "\text{Bayesian update}"']
\end{tikzcd}
$$

Figure 1: Solid arrows denote subjective grounding relations for a given observer: $A \to B$ means every instance of B is physically realised by an instance of A. This paper explores the solid arrows. The dashed arrow is the Diósi–Penrose postulate.

identities on local Rindler horizons but leaves open what physical process underlies each entropy increment at the level of individual microstates; the present paper proposes that process. Van Raamsdonk [3] showed, in the AdS/CFT setting, that entanglement between two spatial regions generates their geometric connection; the present construction realises the same effect in flat Minkowski space, microstate by microstate, with an explicit source. The BMS program [4, 5, 6] identifies the symmetry algebra whose modes are excited at the horizon but does not specify what excites individual modes; the bilocal construction supplies a candidate process.

1.1 Roadmap

Our central technical result is the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$: the Gibbons–Hawking–York boundary term and the matter action count the same measurement events in geometric and informational language respectively, with the Planck area ℓ_P^2 as the conversion factor. Its role here is diagnostic. An identification of microstates that survived only at the thermodynamic ensemble level would be consistent with many underlying microstructures; one that holds event by event is a sharper constraint, and that sharpness is what validates the proposal.

The paper has a two-layer structure. The *heuristic layer* (Section 2) requires only Landauer's principle, the Unruh effect, and Newton's law; it recovers standard black hole thermodynamics from a temperature-free kinematic bound and is self-contained. The *construction layer* (Sections 3–5) is the technically precise core: a Bell state of photons with anticorrelated helicities, sourced by a bilocal term in the action, generates an Aichelburg–Sexl pp-wave at the Rindler horizon; applying the Raychaudhuri equation to the resulting null congruence yields $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ event by event. Sections 6–8 apply this construction to entanglement-generated geometry, the equivalence of Carnot and Unruh–Landauer bounds, and the EFT origin of the GHY term.

2 The Unruh–Landauer Bound and Motivating Heuristics

Two structuring choices run throughout the paper. We work in the non-thermal global Minkowski description rather than the thermal Rindler description: thermality is a derived consequence of restricting to one wedge [7, 8], not a primitive, and individual microstates are non-thermal.

Our approach is a refactored version of the Diósi–Penrose objective collapse program [9, 10], in the sense described in the interpretational stance of Section **??**: measurement is a subjective process mediated by energy–momentum transfer, with state reduction as a

Bayesian update for a given observer grounded in common causal ancestry, as illustrated in Figure 1.

This section has two distinct parts. The Unruh–Landauer relation of Section 2.1 is an exact result that the construction layer inherits directly. substituting the Unruh temperature into Landauer's bound eliminates temperature entirely, leaving a purely kinematic inequality depending only on acceleration. The subsequent derivations are explicitly Verlinde-style heuristics [2]: the same actors appear, acceleration, the Unruh relation, Newton's laws, but thermality is eliminated rather than embraced. Where Verlinde deploys the thermal Rindler frame as the mechanism connecting entropy to force, the Unruh–Landauer relation eliminates temperature entirely, and Newton's laws then recover standard black hole thermodynamics directly from the resulting kinematic bound. Thermality is a consequence of restricted causal access to one Rindler wedge [7, 8], not a primitive.

2.1 The Unruh–Landauer Relation

Landauer's principle [11] establishes k_BT as the natural energy scale of one distinguishable degree of freedom in a thermal environment at temperature T: any physical carrier representing ΔH nats must have at least this energy to be distinguishable from thermal noise,

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates k_BT, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the energy of the physical carrier required to register one nat of information under proper acceleration a; acceleration sets the minimum energy scale of the carrier. The full thermodynamic accounting, where the Landauer erasure cost appears, is deferred to Section 7.

The connection between Landauer's principle and black hole thermodynamics has been noted previously. Cortês and Liddle [12] showed that Hawking evaporation saturates the Landauer bound at the Hawking temperature, identifying the two results as expressions of the same process at the level of bulk evaporation. Related bounds on the energy cost of information acquisition in curved spacetime have been derived using the second law of thermodynamics applied to relativistic communication channels [13]. The present bound (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation, yielding a purely kinematic bound depending only on acceleration, and the bound is applied to individual carrier quanta rather than to bulk evaporation.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $4\pi^2/\ln 2$:

$$\lambda_{\text{sat}} = \frac{4\pi^2}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8\ \mathrm{m\,s^{-2}}$, $\lambda_{\mathrm{sat}} \approx 8$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

Note on method. Sections 2.2–2.5 are Verlinde-style dimensional analysis. They use Newtonian gravity and recover correct results at the Schwarzschild scale by the same coincidence exploited by Verlinde [2] and Bekenstein [14]: the Newtonian inverse-square law with a point mass reproduces the Schwarzschild geometry at the horizon scale, even though it formally breaks down there. This coincidence is specific to the spherically symmetric case; other matter distributions, including the bilocal source of Section 3, or the full Einstein equations, will in general give different results. The construction layer of Sections 3–5 makes no use of these heuristics. We make no attempt to disguise any of this.

In what follows, standard results are *reproduced as heuristic consequences* of the Unruh–Landauer relation and Newton's laws. The energy transfer ΔE implied by the relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction as a *thrust quantum.* For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

A measurement event localized to a spatial scale r is bounded by a causal diamond with temporal extent $\Delta t = 2r/c$. This identification is the heuristic input from which the subsequent results follow, then

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2m_d r}. \tag{6}$$

This is closely related to Verlinde's entropic gravity [2]: Verlinde identifies holographic thrust on information screens with the origin of inertia itself, and Newton's laws are a common ingredient in both constructions. The present approach differs in grounding each entropy increment in a discrete measurement event, the measurement impulse playing the role of Verlinde's holographic thrust, rather than in a coarse-grained thermodynamic gradient.

2.3 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton's laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G\Delta E}{c^2 r^2} \tag{7}$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (6),

$$\frac{\Delta E}{2m_d r} = \frac{G\Delta E}{c^2 r^2}. \tag{8}$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r:

$$r = \frac{2Gm_d}{c^2} \equiv r_s, \tag{9}$$

which is precisely the Schwarzschild radius of the detector mass m_d. The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \tag{10}$$

in agreement with the standard Schwarzschild result, as expected from the Verlinde coincidence. This reproduces the acceleration appearing in the Unruh temperature $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the heuristics.

2.5 Bekenstein–Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [15, 16], the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \le \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot \Delta m_d c^2. \tag{11}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \le \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{12}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [14], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$ — which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

Substituting the horizon condition $a = c^2/R$ directly into the Unruh–Landauer bound (3) yields

$$\Delta H \le \frac{2\pi R\, \Delta E}{\hbar c}, \tag{13}$$

which is precisely the Bekenstein universal entropy bound [17] on the entropy-to-energy ratio for a system of size R. The Unruh–Landauer relation and the Bekenstein bound are therefore the same inequality: the former is its acceleration-frame expression, the latter its position-space expression, and the Rindler horizon is the surface at which the two descriptions coincide. The covariant generalisation of the Bekenstein bound to arbitrary null hypersurfaces is the Bousso bound [18], which bounds entropy on any non-expanding null congruence by the area of its generating surface. The Raychaudhuri derivation of Section 5 operates precisely on such a congruence; the present construction can be read as a microstate-level realisation of the Bousso bound, with each handoff event contributing one Planck-area patch to the light-sheet.

Pesci [19] derived the covariant entropy bound from the Unruh temperature by applying the second law of thermodynamics across local Rindler horizons, the closest precursor to the present argument. The present bound equation (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation into Landauer's principle rather than retained as an input to the second law, yielding a purely kinematic bound depending only on acceleration; and the bound is applied to individual carrier quanta rather than to a bulk thermodynamic flux. The identification of equation (3) with the Bekenstein bound [17] at the horizon condition $a = c^2/R$, and the reading of both as the same inequality in different frames, appears to be new in this form.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation.

The arguments of Section 2 establish a heuristic relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. In [20] this logic was inverted for a scalar field: a source placed in the past injects non-thermal entangled excitations into the field, and the observer's accelerated response was shown to be driveable by these source-induced excitations. We extend that construction here to the spin-1 electromagnetic field, upgrading from scalar to helicity-carrying Bell states and making the single-bit structure of each measurement event explicit.

A *Minkowski microstate* is, in this paper, a single explicit quantum event in flat Minkowski space, a photon pair emitted by a physical source, detected by two accelerated observers, that realises one discrete entropy increment of the kind that Jacobson and Verlinde require at the thermodynamic level. We make no claim that this is the unique such construction; other configurations, with back-reaction, asymmetric detector placement, or different source geometries, would also qualify. The construction below is simply the smallest one we could exhibit that is clean enough to track consistently through the geometric, thermodynamic, information-theoretic, and effective field theory frameworks simultaneously.

3.1 A Source and Two Observers

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ – *Right*, $\mathcal{L}$ – *Left*, and $\mathcal{P}$ – *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced by construction: the two photons carry spatial momenta $+k\hat{z}$ and $-k\hat{z}$ respectively, so the total linear momentum vanishes, $k + (-k) = 0$. The angular momentum vanishes term by term: in each helicity assignment $(\lambda_R, \lambda_L) = (+1, -1)$ and $(-1, +1)$, the total helicity is $\lambda_R + \lambda_L = (+1) + (-1) = 0$ and $(-1) + (+1) = 0$ respectively, and the symmetric Bell state superposition (26) preserves this since both terms contribute zero independently. The source therefore exhibits no net back-reaction in either linear or angular momentum. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively via absorption, the momentum impulse of which accelerates each detector into its respective Rindler wedge; each observer thereby registers a helicity eigenvalue and acquires one classical bit of information. The emission event at $\mathcal{P}$ naturally defines null coordinates $u = t - z + 1$ and $v = t + z + 1$, centered so that $\mathcal{P}$ lies at $u = v = 0$, the right-moving photon propagates along $u = 0$, and the left-moving photon propagates along $v = 0$.

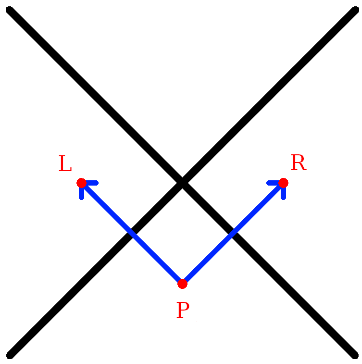

Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ into the right Rindler wedge and $\mathcal{L}$ into the left Rindler wedge. The two null directions of propagation define the null coordinates $u = t - z + 1$ and $v = t + z + 1$, with the right-moving photon propagating along $u = 0$ and the left-moving photon along $v = 0$; these are the z^- and z^+ axes of the double-null coordinate system of Section 3.4.

The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ realises a form of tripartite entanglement with a natural interior reconstruction interpretation. Neither $\mathcal{R}$ nor $\mathcal{L}$ alone can distinguish a correlated Bell-state emission at $\mathcal{P}$ from an uncorrelated thermal source: each observer's reduced density matrix is thermal, exactly as in the standard Unruh effect. Together, their two anticorrelated bits reveal the correlation structure of the emission event, identifying it as a specific Bell state at a specific frequency, and providing a microstate-level analogue of black hole interior reconstruction [21, 22], where the interior is recovered as classical correlations in the joint record of complementary exterior observers.

3.2 PVM Localization and the Elimination of Coarse-Graining

The Unruh mode plays a double role in this construction. As a QFT object it is a coefficient in a mode expansion, a solution to the wave equation analytically continued from the Rindler wedge into all of Minkowski space. But once $\mathcal{P}$ is specified as a real physical source via the squeezing interaction (20), the mode function acquires a second interpretation: it is a probability amplitude in the Born rule sense, encoding where $\mathcal{R}$ is likely to detect the emitted quantum. This is the inversion of the standard Unruh–DeWitt logic. Rather than coupling a detector to a pre-existing acceleration and reading off a thermal distribution by averaging over all Rindler frequencies, we front-load the logic at $\mathcal{P}$: the source fixes the frequency ω_k and the mode function then specifies the probability amplitude for detection.

Thermality, in this picture, is not a primitive property of the vacuum seen by an accelerating observer but a consequence of residual ignorance, about the precise location of $\mathcal{R}$ within the wedge, and about the spatial origin of the detected excitation, that the present construction resolves by specifying the common causal ancestry of $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{L}$ explicitly. The ignorance is epistemic, not ontic: the global Minkowski state remains pure throughout, and thermality is what a single-wedge observer sees when causal access to $\mathcal{P}$'s emission event is unavailable.

This inversion has a geometric consequence that runs through the entire paper: it is precisely what allows the construction to live in flat Minkowski space rather than requiring an AdS/CFT dual. In the standard holographic picture, geometry is primary and entanglement is read off it, Van Raamsdonk's argument [3] proceeds from the entanglement entropy of a boundary state to the geometry of the bulk ER bridge.

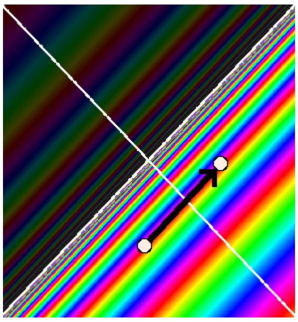

Figure 3: The points $\mathcal{P}$ and $\mathcal{R}$ are marked with white dots, connected by the null geodesic (black arrow). The rainbow coloring shows the Unruh mode phase structure across Minkowski space. A PVM projection onto this null line selects a single frequency from the thermal superposition, replacing the full delocalized mode with the localized non-thermal microstate of equation (26).

Here the direction of explanation is reversed: the emission event at $\mathcal{P}$ is primary, and the geometry,the AS pp-wave, the horizon area increment, the BMS supertranslation, is the *ontic* output of the subsequent measurement events at $\mathcal{R}$ and $\mathcal{L}$. The geometry is a real physical change; what is epistemic is the assignment of that change to a particular observer's microstate record. These are distinct, and keeping them distinct is what allows the construction to live in flat Minkowski space without a holographic dual.

Longitudinal localization. The primary localization is in the longitudinal (t, z) plane, which is the setting of Figure 3 and of the parabolic cylinder function analysis of [20]. In that work, a Minkowski mode φ_q^* multiplied pointwise by a Gaussian shaping function $N_{\mu,\sigma}$ of width σ in (t, z), then paired via the Klein–Gordon inner product with a Rindler mode r_k, yields an overlap expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$. The Gaussian could equivalently have been applied to the Rindler side; what matters is that σ controls the degree of longitudinal localization of the source-detector pair. The Stokes phenomenon in the asymptotics of $D_\nu(-\zeta)$ as σ varies drives a transition between two physically distinct regimes: $\sigma \to \infty$ gives the thermal regime where the $e^{-\frac{1}{4}\zeta^2}$ branch dominates, and $\sigma \to 0$ gives the localized non-thermal regime where the $e^{+\frac{1}{4}\zeta^2}$ branch dominates, with the Stokes line at $\arg\zeta = \pi/4$ marking the boundary between them.

In the present paper the same Gaussian shaping function is reread as a Kraus operator acting on the quantum state.

$$K_\sigma = \int e^{-u^2/2\sigma^2}\; |u\rangle\langle u|\; du, \tag{14}$$

where $u = t - z + 1$ is the null coordinate[1] defined in Section 5. The operator K_σ together with its complement

$$K_\sigma^\perp = \int \sqrt{1 - e^{-u^2/\sigma^2}}\; |u\rangle\langle u|\; du, \tag{15}$$

satisfying $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ by construction, defines a one-parameter POVM family; the Stokes transition between its two asymptotic regimes is illustrated in Figure 4. Specifying $\mathcal{R} = (0, 1)$ and $\mathcal{P} = (-1, 0)$ is precisely the $\sigma \to 0$ limit: the Kraus operator sharpens to a PVM projector onto the unique null geodesic connecting them, selecting a single definite frequency ω_k from what would otherwise be a thermal superposition over all Rindler frequencies. This is visible directly in Figure 3: the rainbow phase structure of

[1] We use ζ for the Stokes variable of [20] to avoid confusion with the longitudinal spatial coordinate z.

the Unruh mode concentrates onto the null line between $\mathcal{P}$ and $\mathcal{R}$, and the PVM projection selects that line. The Stokes transition of [20] is therefore not merely a field-theoretic regularization result but the analytic signature of the POVM-to-PVM transition: the same mathematical object is being read in two complementary languages, field theory and measurement theory, with the Stokes line marking the boundary between them. The helicity label λ enters the longitudinal mode function as a spectator index and does not affect the u-dependence of the Kraus operator (14); the Stokes transition analysis of [20] therefore applies independently to each helicity sector without modification.

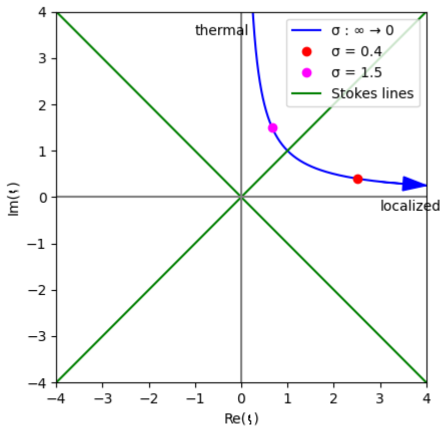

Figure 4: Trajectory of $\zeta = iq\sigma + \mu/\sigma$ in the complex plane as σ interpolates between the two limits of the Kraus family (14). The thermal limit $\sigma \to \infty$ corresponds to ζ on the positive imaginary axis; the localized PVM limit $\sigma \to 0$ corresponds to ζ on the positive real axis. The green Stokes lines at $\arg \zeta = \pi/4$ mark the boundary between the two asymptotic regimes of the parabolic cylinder function $D_\nu(-\zeta)$, and therefore the boundary between the POVM and PVM descriptions of the longitudinal localization. Figure from [20].

Transverse localization. The transverse (x, y) directions require a separate and independent localization. A Rindler plane wave at frequency ω is a coherent superposition over all possible source-detector pairs $(\mathcal{P}', \mathcal{R}')$ connected by null geodesics at that frequency, uniform in the transverse plane. The longitudinal and transverse localizations decouple because the wave equation in double-null coordinates separates: the u-dependence and the (x, y)-dependence of the mode function enter independently, so the two Kraus projections can be applied in either order without interference. Specifying $\mathcal{P} = (-1, 0)$ and $\mathcal{R} = (0, 1)$ performs a second PVM projection, now in the transverse directions, selecting the unique null geodesic and resolving the remaining coarse-graining over transverse position. The transverse shaping is controlled by an independent width $\sigma_\perp$, which we treat as a free parameter in Section 3.7 and whose physical consequences are developed in Section 5.

The two widths σ and $\sigma_\perp$ control physically distinct aspects of the construction and need not be sent to zero together. The longitudinal σ governs the POVM-to-PVM transition and the thermal-to-non-thermal transition analyzed via the Stokes phenomenon; Section 3 works entirely in the $\sigma \to 0$ PVM limit. The transverse $\sigma_\perp$ controls the spatial profile of the horizon perturbation and the BMS mode content: $\sigma_\perp \to \infty$ produces a planar impulsive wave exciting a single BMS supertranslation mode, while $\sigma_\perp \to 0$ recovers the Aichelburg–Sexl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$ exciting the full BMS spectrum. The central results of the paper, the area increment $\Delta A = 8\pi G\hbar\omega$ and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$, are independent of $\sigma_\perp$, fixed by energy conservation alone.

The microstate. The result of both projections is an individually non-thermal microstate carrying a definite Rindler frequency ω and a definite longitudinal position on the null geodesic from $\mathcal{P}$ to $\mathcal{R}$. Thermality is not a property of any single microstate but of the ensemble; it emerges when one models the detector response within a wedge as a sum over all Rindler modes, as in the standard Unruh–DeWitt treatment [15, 25], and is a consequence of that mode-averaging rather than a premise of the construction. The individual non-thermal microstate is what the PVM projection onto the $\mathcal{P}$–$\mathcal{R}$ null geodesic selects; the thermal ensemble is what remains when neither coarse-graining is resolved and $\mathcal{L}$ is traced out.

A note on the $\sigma \to 0$ idealization: throughout this paper the PVM limit is understood as the idealization of a physical construction that lives at small but finite σ. At exactly $\sigma = 0$ the Kraus operator (14) becomes a projector onto a set of measure zero in the continuum $|u\rangle$ basis and is no longer normalizable in the strict sense; similarly, the AS longitudinal shift $\Delta v \propto -\ln r^2$ diverges on axis at $r = 0$ in the strict localized limit. Both divergences are regularized by keeping $\sigma > 0$, as carried out explicitly for the geometric side in Appendix 11. The parameter σ therefore plays a unified role: it is simultaneously the width of the Kraus operator in the longitudinal direction, the transverse profile width of the photon mode, and the regularization scale of the AS singularity, all controlled by the same physical quantity, the photon wavelength $\sigma \sim c/\omega$. All sharp statements about individual microstates should be understood as holding in the limit $\sigma \to 0^+$ rather than at exactly $\sigma = 0$.

3.3 Helicity Structure and PCT Symmetry

We now upgrade this construction from a spin-0 scalar field in [20] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the PCT symmetry between wedges, which by the Bisognano–Wichmann theorem [8] is implemented by the modular conjugation operator J mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

The anticorrelated helicity assignment traces to the PCT symmetry implemented by the Bisognano–Wichmann modular conjugation J, which maps the right-wedge algebra to the left-wedge algebra and simultaneously flips $\lambda \to -\lambda$; the Bell state (26) is the unique state in this momentum sector invariant under that operation. The common algebraic origin of this anticorrelation in the Lorentz algebra decomposition $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$, and its connection to the "tick-tock" mechanism of Section 4.7, is developed in detail in Section 4.7.

3.4 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$, with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^\pm = \frac{1}{\sqrt{2}}(t \pm z) = \frac{1}{\sqrt{2}}(u - 1 \pm (v - 1)), \tag{16}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Note that $z^\pm$ are centered on the Minkowski origin $(t, z) = (0, 0)$ rather than on $\mathcal{P}$; the coordinates u, v defined in Section 3 are the $\mathcal{P}$-centered versions and are used throughout the remainder of the paper. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0, \tag{17}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the PCT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$; consistency with Maxwell's equations and the absence of over-determination are verified in Appendix 10.

The lack of manifest four-dimensional covariance in equation (17) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k^\mu_R, k^\mu_L\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice: the helicity eigenvalues $\lambda = \pm 1$ are representations of the massless little group $ISO(2)$ and are therefore Poincaré-covariant objects independent of how the gauge is fixed, and the consistency of the double-null gauge with Maxwell's equations and the correct degree-of-freedom count is verified explicitly in Appendix 10.

3.5 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x, y) = g \left(\epsilon^{i*}_{(-1)}\, \epsilon^{j}_{(+1)} + \epsilon^{i*}_{(+1)}\, \epsilon^{j}_{(-1)} \right) f_L(x)\, f_R(y), \tag{18}$$

where f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{19}$$

The two terms in (18) are related by the PCT transformation in the sense of Bisognano–Wichmann [8], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \to -\lambda$.

More precisely, the modular conjugation J of Bisognano–Wichmann maps a right-wedge creation operator $a^{R\dagger}_{k,(\lambda)}$ to a left-wedge creation operator $a^{L\dagger}_{-k,(-\lambda)}$, implementing PCT between wedges. The Bell state (26) is a natural two-photon state at fixed frequency ω and fixed momentum pair $(k, -k)$, invariant under J: within this momentum sector, the anticorrelated helicity assignment and the symmetric superposition are not choices but consequences of requiring J-invariance, since J flips both helicity and wedge simultaneously, leaving no room for a relative phase or helicity-correlated alternative. The bilocal source (18) is therefore a natural source, consistent with the algebraic symmetry of the Rindler vacuum, and the simplest one we could construct with the required properties.

The symmetrization is consistent with PCT symmetry and angular momentum conservation at the emission vertex. Other source geometries are possible; this one is chosen for its simplicity and the transparency of its connection to the algebraic structure of the Rindler vacuum. The argument has three steps. First, a scalar source carries zero angular momentum, so the emitted pair must satisfy $\lambda_R + \lambda_L = 0$, which forces the two helicities to be anticorrelated: the only possibilities are $(\lambda_R, \lambda_L) = (+1, -1)$ or $(-1, +1)$.

Second, the Bisognano–Wichmann modular conjugation J maps the first assignment to the second and vice versa, so the unique J-invariant state in this sector is their symmetric superposition with relative phase fixed by J-invariance. Third, any other relative phase or asymmetric combination would break the PCT symmetry between wedges, which is a symmetry of the Minkowski vacuum and must therefore be preserved by the source. The total angular momentum vanishes in both terms independently.

The bilocal source selects a specific microstate from the Minkowski vacuum by a PVM projection onto the $n = 1$ sector at fixed ω and fixed helicity pair $(\lambda, -\lambda)$, in exact analogy with the scalar construction of [20]. No perturbative expansion in a coupling constant is required: the helicity-anticorrelated photon pair is already present in the vacuum decomposition (24), and the bilocal source selects it via the PVM projection of Section 3.2. The operator content of this selection is

$$a^{L\dagger}_{-k,(-1)}\, a^{R\dagger}_{k,(+1)} \;+\; a^{L\dagger}_{-k,(+1)}\, a^{R\dagger}_{k,(-1)} \;+\; \text{h.c.}, \tag{20}$$

where the two creation operator pairs correspond to the two PCT-related helicity assignments, each carrying total angular momentum zero, consistent with emission from a scalar source at $\mathcal{P}$. This is the vector-field analogue of the scalar squeezing interaction [26], with the PCT symmetry between wedges made explicit; the connection to the quantum optics squeezing literature is structural rather than perturbative.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate. The acceleration a does not fix the frequency ω of each selected quantum; rather, it determines the rate $\dot{N}$ at which the bilocal source selects entangled pairs in order to implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome, one bit in the sense of one binary alternative carrying $\ln 2$ nats, and these are paired by the construction independently of a. The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

3.6 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(z) = \int_0^\infty \frac{d\omega}{2\pi}\,\frac{1}{\sqrt{2\omega}}\,\epsilon^i_{(\lambda)}\left[b^R_{\omega,\lambda}\,u^R_\omega(z) + b^{R\dagger}_{\omega,\lambda}\,u^{R*}_\omega(z)\right], \tag{21}$$

where $u^R_\omega(z)$ are the right-wedge Rindler modes at Rindler frequency ω, and $b^R_{\omega,\lambda}$, $b^{R\dagger}_{\omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[b^R_{\omega,\lambda},\, b^{R\dagger}_{\omega',\lambda'}\right] = \delta(\omega-\omega')\,\delta_{\lambda\lambda'}. \tag{22}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency ω, the Unruh modes are

$$c_{\omega,\lambda} = \cosh\theta_\omega \; b^R_{\omega,\lambda} - \sinh\theta_\omega \; b^{L\dagger}_{\omega,-\lambda}, \tag{23}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a. The helicity flip $\lambda \to -\lambda$ on the left-wedge operator is not an independent assumption but a direct consequence of the Bisognano–Wichmann modular conjugation J, which acts as $J\, b^{R\dagger}_{\omega,\lambda}\, J^{-1} = b^{L\dagger}_{-\omega,-\lambda}$ and therefore requires the left-wedge partner of a right-wedge mode of helicity λ to carry helicity $-\lambda$; the PCT symmetry between wedges is thereby reflected in the superposition over both helicity assignments in the Bell state. The transformation (23) is the standard Bogoliubov relation for a massless field [15, 27], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\omega} \sum_{n=0}^{\infty} \tanh^n\theta_\omega \; |n_{\omega,\lambda}\rangle_R \, |n_{\omega,-\lambda}\rangle_L, \tag{24}$$

and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle^{(\lambda)}_M. \tag{25}$$

Equation (24) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source specifies a physical emission event: exactly one pair, at fixed ω, with fixed helicity assignment. The PVM projection of Section 3.2 identifies this event with the $n = 1$ term in the vacuum decomposition (24) at that ω and helicity pair. The higher n terms are not suppressed by any weak-coupling assumption; they correspond to different physical events — two-pair emissions, three-pair emissions, and so on — that this source simply does not describe. The selection is prescriptive, not perturbative. The resulting single microstate is

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left(|1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L \right). \tag{26}$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (23) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. Thermality, when it arises from coarse-graining over the ensemble of microstates (26), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

3.7 Two Geometric Regimes

The microstate (26) admits two complementary geometric descriptions depending on the transverse localisation of the photon mode, corresponding to the two limits of the transverse Kraus family

$$K_{\sigma_\perp} = \int e^{-r^2/2\sigma_\perp^2} \; |\mathbf{x}_\perp\rangle\langle\mathbf{x}_\perp| \; d^2x_\perp, \tag{27}$$

parametrized by $\sigma_\perp$, the direct transverse analogue of the longitudinal Kraus family (14) with the null coordinate u replaced by the transverse radial coordinate $r = |\mathbf{x}_\perp|$. The bilocal source fixes a definite Rindler frequency ω but imposes no constraint on transverse momentum. A mode with zero transverse momentum is uniform across the horizon cross-section; a wave packet of characteristic width $\sigma \sim c/\omega$ is localised within one wavelength of the photon axis. These two limits correspond to geometrically distinct pp-waves at the Rindler horizon, both analysed in Section 5:

- **Delocalized limit** ($\sigma_\perp \to \infty$, POVM $\to \mathbf{1}$): the stress tensor is uniform across the horizon, producing a *planar impulsive wave* with longitudinal shift $\Delta v \propto -r^2$. Exactly one BMS supertranslation mode is excited, because the constant-Laplacian supertranslation field sourced by a uniform stress tensor corresponds to a single $\ell = 2$, $m = 0$ solid harmonic in the angular mode decomposition; the full argument is given in Section 5.1.
- **Localized limit** ($\sigma_\perp \to 0$, POVM $\to$ PVM): the stress tensor concentrates near the beam axis, recovering the Aichelburg–Sexl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$. All BMS supertranslation modes are excited.

Both limits produce the same total horizon area increment $\Delta A = 8\pi G\hbar\omega$, as shown in Section 5 and Appendix 11. The physical reason is that the Raychaudhuri equation integrates the null energy T_{uu} over the entire transverse horizon cross-section: regardless of how that energy is distributed transversely, the integral $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$ is fixed by energy conservation alone, so the total area increment is insensitive to the transverse profile. The difference between the two limits lies entirely in the spatial distribution of that increment across the horizon and the corresponding BMS mode content excited by each profile.

4 Matter as a Substrate

This section develops the physical picture underlying the bilocal construction of Section 3: matter as an information-processing substrate, and the horizon as the surface at which that substrate's descriptive capacity saturates. The section has three interlocking threads.

The first thread (Sections 4.1–4.2) is a channel-theoretic argument: a massive particle carries phase at the Compton frequency mc^2/h, which simultaneously sets its clock rate, its sampling rate via Bremermann's limit, and its maximum information throughput. The horizon enters as the surface at which this timelike channel hands off to the spacelike horizon channel, with the Planck area ℓ_P^2 as the conversion factor between the two.

The second thread (Sections 4.3–4.4) is a formal definition and its consequences: the handoff event is defined precisely as the transfer of one quantum of action $\hbar$ and one classical bit across the horizon, and the channel capacity is shown to saturate at the Rindler horizon at exactly one bit per Rindler period, the same condition as the PVM limit of Section 3.2 and the EFT boundary of Section 8.

The third thread (Sections 4.6–4.7) broadens the picture: the distinction between massless carriers and massive memory substrates is shown to be the operational content of the mass/massless divide in the Standard Model, and the tick-tock mechanism connects the horizon handoff to the Penrose zig-zag construction for the Dirac equation, making the bilocal Bell state and the Dirac 4-spinor instances of the same PCT-invariant structure at different codimension.

The two results that Section 5 uses directly are the handoff definition of Section 4.3 and the saturation condition of Section 4.4; the remaining material is conceptual context that motivates and connects these results to the broader framework.

This viewpoint sidesteps a prior question about quantum gravity: how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom? Common approaches posit Planck-scale cells [33] or discrete spacetime events [34], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [35]. We instead derive discreteness from matter: space and time are properties of matter itself, defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

The identification of proper time with accumulated action connects to two independent programs in the literature. The Page–Wootters mechanism [30, 31] derives time evolution from entanglement between a clock system and the rest of the universe, consistent with the view that proper time is an emergent rather than background quantity. The modular flow perspective of Connes and Rovelli [32] gives a complementary algebraic picture: in a generally covariant quantum theory the physical flow of time is the modular flow of the state with respect to the observable algebra, which the Bisognano–Wichmann theorem identifies with Rindler time evolution for the vacuum restricted to one wedge. Both programs support the view that proper time is emergent; the present construction does not depend on either but is consistent with both.

4.1 Matter as a Timelike Channel

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/h$ is the natural clock rate of a mass m, and the Bohr–Sommerfeld quantization condition $\oint p\,dq = nh$ is its coarse-grained expression: it is the correct effective description at energy scales below mc^2, where the sub-Compton phase structure is unresolved and action accumulates in integer units of h. This is not a limitation but a feature: the Bohr–Sommerfeld condition holds exactly within the EFT window defined by the Compton scale as UV cutoff and the horizon distance as IR cutoff (Section 8), where the Heisenberg uncertainty $\Delta E \Delta t \sim \hbar$ prevents resolution of sub-Compton phase structure and action accumulates in integer units of h to within the precision allowed by that window. Bremermann's computational limit [28] independently recovers the same rate as the maximum clock rate of any physical system with energy E, and the two agree because $\hbar$ is the minimal unit of physical evolution at that scale.

Each quantum corresponds to one sample[2] of the phase $e^{iS/\hbar}$ in the path integral. The matter substrate gives rise to a channel capacity in the Shannon sense: for a system of mass m with internal Hilbert space dimension d, the maximum rate of information flow is

$$C = \underbrace{\frac{mc^2}{h}}_{\text{bandwidth}} \times \underbrace{\log_2(d)}_{\text{dynamic range}} \qquad \text{(bits per second)}, \tag{28}$$

where the bandwidth is the Bremermann sampling rate and the dynamic range is the information content per sample. Both quantities are extensive under composition:

$$\frac{mc^2}{\hbar} = \frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \tag{29}$$

$$\log_2(d) = \log_2(d_1 d_2) = \log_2(d_1) + \log_2(d_2), \tag{30}$$

consistent with their interpretation as properties of matter that compose naturally under union. The signal being sampled at the Bremermann rate is the internal quantum state of the particle, its spin, helicity, and other quantum numbers, and the dynamic range is $\log_2$ of the dimension of the internal Hilbert space. The channel capacity (28) sets the rate at which matter can accumulate and process information, under the assumption that each Bremermann sample accesses the full d-dimensional internal Hilbert space; for composite systems the accessible dimension per sample may be less than d, in which case (28) is an upper bound rather than an equality. For the elementary particles of Section 3, where $d = 2$ corresponds to the two helicity eigenstates, the bound is saturated by construction. Section 4.4 shows that this rate saturates at the Rindler horizon, making the horizon the natural IR boundary of the matter channel and anticipating the EFT breakdown of Section 8.

[2] We use the information theoretic term *sample* throughout to denote a single zero crossing of the phase $e^{iS/\hbar}$ in the path integral. The term *event* or *tick* may be substituted throughout. It is defined only up to the Heisenberg resolution.

4.2 The Horizon as a Spacelike Channel

The three fundamental constants c, $\hbar$, and G play precise roles in the channel picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the timelike channel via the Compton frequency $mc^2/\hbar$. The gravitational constant G, through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action.

The horizon enters the channel picture as the surface at which the matter-memory on one side becomes causally disconnected from the matter-memory on the other. The photon passes through the horizon channel and deposits into the matter channel on the other side; the handoff is the central event of the construction.

The Bekenstein and Bremermann bounds count the same physical event from orthogonal directions, related by the Unruh–Landauer relation of Section 2. For a single photon of dimensionless Rindler frequency $\Omega \equiv \omega/a$ crossing the horizon, the Bremermann count is the number of new resolution elements added to the receiving timelike channel per Rindler period. The Rindler period $\tau = 2\pi/a$ is the natural time unit of the accelerated observer: it is the period of the thermal oscillations seen by a Rindler detector at acceleration a, set by the Unruh temperature via $\tau = \hbar/k_B T_H = 2\pi/a$ in natural units. The Bremermann count per Rindler period is then:

$$\Delta f \cdot \tau = \frac{\omega}{2\pi} \cdot \frac{2\pi}{a} = \Omega. \tag{31}$$

That the Bekenstein spatial count agrees with this Bremermann temporal count, up to the universal Rindler factor 2π tracing to the analytic continuation between Minkowski and Rindler coordinates, is a consequence of the central result $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$, established geometrically in Section 5; we defer the proof to that section.

4.3 The Handoff Event

The photon crossing the horizon is the handoff between the timelike matter channel and the spacelike horizon channel. We now define this object precisely, so that Section 5 can identify it with a specific geometric structure rather than deriving a coincidental equality.

Definition (Handoff Event). A *handoff event* at a horizon surface $\mathcal{H}$ consists of:

(i) a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$, selected by the PVM projection of Section 3.2, transferring one quantum of action $\hbar$ across $\mathcal{H}$;

(ii) the receiving timelike channel incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$, as established by the Bremermann count (31) of Section 4.2;

(iii) the helicity outcome λ recorded as one classical bit in the receiving channel's matter-memory; and

(iv) the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the handoff explicit.

Each handoff event has exactly two irreducible constituents: one quantum of action $\hbar$ written into matter-memory, incrementing the Bremermann sampling rate, and one bit of classical information, the helicity outcome. These are not independent: they are the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon, two properties of the same physical object that cannot be separated. With this definition in hand, the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5 becomes a counting statement: the GHY boundary term sums $\hbar\omega$ over handoff events because that is what the term geometrically records, one contribution per event.

4.4 Channel Capacity Saturation at the Horizon

The saturation wavelength $\lambda_{\text{sat}} = (2\pi/\ln 2)(c^2/a)$ of Section 2 acquires a precise meaning in the channel picture. A photon of frequency ω absorbed by a detector increments its Bremermann sampling rate by $\Delta f = \omega/2\pi$. Over one Rindler period $\tau = 2\pi/a$, the number of new resolution elements added to the timelike channel is

$$\Delta f \cdot \tau = \frac{\omega}{2\pi} \cdot \frac{2\pi}{a} = \frac{\omega}{a} = \Omega. \tag{32}$$

The Landauer saturation frequency follows from $\lambda_{\text{sat}} = 2\pi/\omega_{\text{sat}}$:

$$\omega_{\text{sat}} = a \ln 2. \tag{33}$$

Substituting into (32):

$$\Delta f \cdot \tau\big|_{\omega=\omega_{\text{sat}}} = \frac{\omega_{\text{sat}}}{a} = \ln 2. \tag{34}$$

The Landauer-saturating photon adds exactly $\ln 2$ nats, one bit, of Bremermann channel capacity per Rindler period. A photon below saturation ($\omega < \omega_{\text{sat}}$) adds less than one bit per period and cannot constitute a complete measurement record; a photon above saturation adds more than one bit and deposits more than the minimum cost of a handoff event. The horizon is therefore the surface at which the minimum energy cost of information acquisition (Landauer) and the minimum time cost of processing it (Bremermann) coincide in a single-bit handoff. This saturation condition is also the natural boundary of EFT validity: the six equivalent conditions of Proposition 1 all reduce to $\omega = \omega_{\text{sat}}$ at the horizon, so the EFT breaks down exactly where the channel saturates.

This saturation condition coincides with the PVM limit of the longitudinal Kraus family (14): the diagonal Kraus operator becomes a sharp projector exactly when the photon wavepacket carries one Landauer bit per Rindler period, connecting the information-theoretic saturation to the geometric localization of Section 3.2.

4.5 Lorentz Covariance

The discreteness of the channel picture, one quantum of action $\hbar$ per handoff event, one Planck-area patch per bit, might appear to conflict with Lorentz invariance, since a preferred frame seems required to count events. This conflict does not arise. The physical events of the construction, the emission at $\mathcal{P}$, the photon pair, the AS pp-wave, the area increment $\Delta A = 8\pi G\hbar\omega$, are all Lorentz-covariant objects. Helicity is a representation of the massless little group and is Poincaré-covariant; the AS pp-wave is a solution to the diffeomorphism-covariant Einstein equations; the Raychaudhuri equation and the GHY boundary term are diffeomorphism-invariant. What is observer-relative is the *decomposition* of these events into a particular Rindler mode basis: a boosted Rindler observer assigns a different frequency ω' to the same photon, but the resulting area increment $\Delta A = 8\pi G\hbar\omega'$ is the same physical event viewed in a different frame, with the total Bekenstein–Hawking count preserved under the boost.

This contrasts with approaches that discretise spacetime geometry directly, Planck-scale cells [33] or causal sets [34], where Lorentz invariance is recovered only statistically at the macroscopic level rather than holding event by event. In the present construction, discreteness lives in the matter sector ($\hbar$ is a Lorentz scalar; photons are covariant objects), and the geometry inherits it through the covariant Raychaudhuri equation. No statistical recovery is required.

4.6 Types of Particles and Interactions

The distinction between massless and massive degrees of freedom is operational in the channel picture: massless quanta transmit information between matter substrates as handoff carriers, while massive systems retain it as matter-memory. A massless particle has no rest frame, no proper time, and carries a single quantum of action between emission and absorption; it does not sample and therefore carries no persistent memory. Its role is precisely that of a pure information carrier effecting a handoff between the timelike channel of one massive body and that of another.

A single photon is the minimal carrier because it contributes exactly one of each:

- **One quantum of action** $\hbar$: the physical transfer, deposited as energy $\hbar\omega$ into the detector, incrementing its Bremermann sample rate by $\Delta f = \hbar\omega/h$.
- **One bit of polarization**: the informational outcome, the classical record of which helicity $\lambda = +1$ or $\lambda = -1$ was absorbed, persisting in the detector's state until erased at a minimum cost governed by Landauer's principle.

These two constituents are not independent: the action quantum $\hbar$ and the helicity $\lambda = \pm 1$ are two aspects of the same object, the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon.

From a given observer's perspective, interactions fall into three classes. **Absorption** produces a persistent classical record: a quantum of action is deposited and information is written. **Emission** is the reverse: stored information is released and the record is lost. **Reflection** is absorption followed by emission; whether it constitutes an information-bearing interaction depends on whether a persistent record survives. The ideal mirror of Section 7 is precisely the limiting case of reflection with no persistent record: it absorbs the photon, temporarily stores the helicity outcome, and re-emits without retaining a copy, making the interaction thermodynamically reversible at the cost of paying the Landauer erasure price at re-emission.

These categories are observer-relative in the same sense as quantum state reduction: just as state reduction is a Bayesian update for a given observer grounded in their causal access to the measurement outcome, the classification of an interaction as absorption, emission, or reflection depends on whether the observer in question retains a persistent record of the outcome. The same physical process may be information-bearing for one observer and purely external for another, depending on whether a record is written into their matter-memory.

4.7 The Tick-Tock Mechanism

The channel picture is not an ad hoc construction: the handoff event of Section 4.3 is the Rindler-horizon version of what Penrose calls the zig-zag construction [40], which we term the tick-tock mechanism to emphasise the alternating rhythm of handoff events rather than the geometry of the path. At each vertex the helicity flips, and it is this alternating helicity that produces spin, visible in Figure 5 as the green loops representing the two $\mathfrak{su}(2)$ factors cycling between handoff events.

The tick-tock is, in precise terms, the bilocal construction of Section 3 realised along a single massive worldline rather than across a spacelike horizon. The Higgs field plays the role of the bilocal source $\mathcal{P}$; the two Weyl spinors of opposite helicity play the role of the two photons propagating to $\mathcal{R}$ and $\mathcal{L}$; each Higgs interaction is a handoff event in the sense of Section 4.3, depositing one quantum of action $\hbar$ and producing one sample of proper time rather than one increment of horizon area. The Bremermann rate mc^2/h is the Higgs interaction rate; the bilocal construction is its (3+1)-dimensional horizon counterpart.

The structural parallel with the bilocal construction and the handoff definition is exact: the two Weyl spinors correspond to the two photons emitted by $\mathcal{P}$; the Higgs interaction

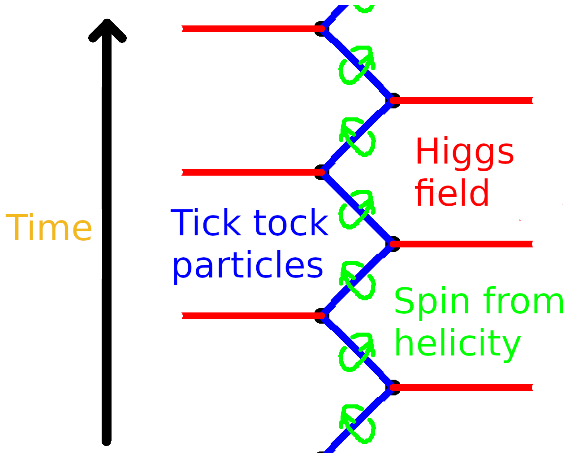

Figure 5: The tick-tock particle picture of the Dirac equation, after Penrose [40]. A massive electron (blue) alternates between right- and left-helicity massless segments, coupled at each vertex by the Higgs field (red). The green loops indicate spin emerging from the alternating helicity, the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ made visual. Each vertex is a handoff event in the sense of Section 4.3: one quantum of action deposited, one helicity outcome recorded, one sample of proper time produced. Adapted from [40]. Penrose's interpretation of the Dirac equation and the Higgs mechanism for an electron.

corresponds to the absorption events at $\mathcal{R}$ and $\mathcal{L}$; between interactions each spinor is massless and moves at c, a pure carrier. Each Higgs interaction is a handoff event in the sense of Section 4.3: it deposits one quantum of action $\hbar$, produces one sample of proper time, records one helicity outcome, and increments the matter-memory of the particle by one Bremermann sample.

The anticorrelated helicity assignment in both constructions has a common algebraic origin in the Lie algebra decomposition already noted above. The Bisognano–Wichmann modular conjugation J acts on right-wedge photon creation operators as

$$J\, a^{R\dagger}_{k,\lambda}\, J^{-1} = a^{L\dagger}_{-k,-\lambda}, \tag{35}$$

flipping helicity and reversing the spacelike separation direction $R \leftrightarrow L$. The PCT operator Θ acts on a right-handed Weyl spinor as

$$\Theta\, \psi_R(t, \mathbf{x})\, \Theta^{-1} \propto \psi_L(-t, -\mathbf{x}), \tag{36}$$

flipping helicity and reversing the timelike separation direction $t \to -t$. In both cases the operation is the same at the representation-theoretic level: exchange of the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$, which is what helicity-flip means algebraically. The operators J and Θ act on different Hilbert spaces and are not equal; what they share is that they both select the same class of invariant configuration, anticorrelated helicities, because PCT-invariance forces that choice independently of whether the separation between the two helicity sectors is spacelike (Rindler horizon) or timelike (Compton worldline). The bilocal Bell state and the Dirac 4-spinor are therefore the same PCT-invariant structure at different codimension, and the Lie algebra is the common home of both.

Mass emerges from the regularity of these interactions: a particle is massive because it undergoes repeated Higgs interactions at the Compton frequency, and its mass is set by that rate via $m = h\dot{N}_{Higgs}/c^2$, where $\dot{N}_{Higgs}$ is the Higgs interaction rate per unit proper time. This is the handoff-event reading of $E = \hbar\omega$: mass is not a static property but a rate, the frequency at which the particle exchanges one quantum of action with the Higgs field. Proper time along a worldline is then a foliation by discrete handoff events, each separating the causal past from the causal future of that interaction, the one-dimensional analogue of the Rindler horizon, and matching the holographic screen picture of Verlinde [2]. The one-dimensional handoff of the zig-zag and the horizon handoff of the bilocal construction are therefore the timelike and spacelike faces of the same elementary event: a PCT-invariant exchange of one quantum of action between two massless helicity eigenstates.

5 pp-Wave Geometry and Action

WLOG we introduce coordinates for the photon traveling from $\mathcal{P}$ $(-1,0)$ to $\mathcal{R}$ $(0,1)$. We work in the null coordinates u, v defined in Section 3, where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and (x, y) are transverse coordinates. Then $(u, v) = (0, 0)$ is $\mathcal{P}$'s location, $(u, v) = (0, 2)$ is $\mathcal{R}$'s location, and $(u, v) = (0, 1)$ is the intersection of the photon with the past right Rinder horizon. The situation is reversed for the left moving photon; we swap u and v to describe photon moving from $\mathcal{P}$ to $\mathcal{L}$.

5.1 The Planar Impulsive Wave: Delocalized Limit

When the photon carries zero transverse momentum its energy is uniform across the horizon cross-section. The stress tensor at the horizon is

$$T_{uu}(u, \mathbf{x}_\perp) = \frac{\hbar\omega}{A_\perp}\,\delta(u), \tag{37}$$

where $A_\perp$ is the transverse horizon area and $\delta(u)$ encodes the impulsive character of the interaction. The corresponding gravitational perturbation is a planar impulsive wave in Brinkmann form

$$ds^2 = -2\,du\,dv + H_{ij}(u)\,x^i x^j\,du^2 + d\mathbf{x}_\perp^2, \tag{38}$$

where the Einstein equation $R_{uu} = 8\pi G T_{uu}$ with $R_{uu} = -\frac{1}{2}\mathrm{tr}(H_{ij})$ gives

$$H_{ij}(u) = -\frac{8\pi G\hbar\omega}{A_\perp}\,\delta(u)\,\delta_{ij}. \tag{39}$$

Area increment. The Raychaudhuri equation on the null congruence gives

$$\frac{d\theta}{du} = -8\pi G\,T_{uu} = -\frac{8\pi G\hbar\omega}{A_\perp}\,\delta(u), \tag{40}$$

so the expansion receives an impulsive kick $\Delta\theta = -8\pi G\hbar\omega/A_\perp$ at $u = 0$. Integrating over the horizon area:

$$\boxed{\Delta A = A_\perp\,|\Delta\theta| = 8\pi G\hbar\omega,} \tag{41}$$

identical to the AS result derived in Section 5.4, here distributed uniformly rather than concentrated near the beam axis.

Longitudinal shift and BMS structure. The gravitational memory of the planar wave is

$$\Delta v(\mathbf{x}_\perp) = \tfrac{1}{2}\bar{H}_{ij}\,x^i x^j = -\frac{4\pi G\hbar\omega}{A_\perp}\,r^2, \tag{42}$$

where $\bar{H}_{ij} = \int H_{ij}\,du$. This quadratic shift satisfies $\nabla_\perp^2(\Delta v) = -8\pi G\hbar\omega/A_\perp = \text{const.}$ A constant-Laplacian supertranslation field corresponds to a single $\ell = 2$, $m = 0$ solid harmonic in the angular mode decomposition: exactly one BMS mode is excited. Compare the AS shift $\Delta v \propto -\ln r^2$, whose Laplacian is $\nabla_\perp^2(\Delta v) \propto \delta^{(2)}(\mathbf{x}_\perp)$; a point-source Laplacian couples to all angular harmonics, so the AS wave populates the entire BMS spectrum.

Cut-and-paste structure. Despite its delocalized transverse profile, the planar impulsive wave retains the cut-and-paste structure of the AS pp-wave: the geometry is flat for $u < 0$ and $u > 0$, joined at $u = 0$ by the identification (42) together with a transverse velocity kick $\Delta\dot{x}^i = \bar{H}^i{}_j x^j = -(8\pi G\hbar\omega/A_\perp)\, x^i$ directed inward. The planar wave is therefore the global, uniform-expansion version of the same horizon surgery that the AS wave performs locally.

5.2 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{43}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta\mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{44}$$

   directed toward the axis. Test particles are pulled toward the centre of the wave, regardless of their initial transverse separation.

5.3 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $-v$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 6.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx\, dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

5.4 Transverse Focusing Increments Horizon Area

The Raychaudhuri calculation of Section 5.1 already establishes $\Delta A = 8\pi G\hbar\omega$ in the delocalized limit via equation (41). We now derive the same result for the localized AS pp-wave, confirming that the area increment is independent of the transverse localisation of the photon.

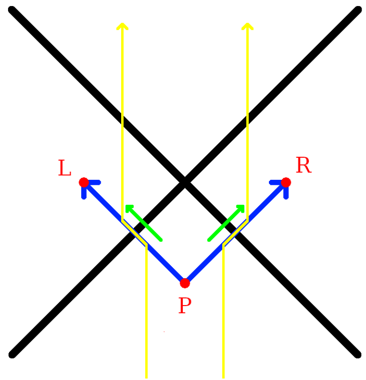

Figure 6: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

The transverse focusing of the AS pp-wave produces a calculable increment in horizon area. We derive this using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1] and consistent with the generalized second law for Rindler horizons proved by Wall [41], but applied here to a single microstate rather than a thermodynamic or ensemble average.

For a null congruence with generators k^μ and expansion θ, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{45}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp. \tag{46}$$

In a strict AS construction, the source is a point particle and $T_{\mu\nu}$ is distributional, concentrated on the axis. The present construction provides a natural regularization: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

Substituting into equation (46) gives

$$\boxed{\Delta A = 8\pi G\, \hbar\omega = 8\pi G\, \Delta E.} \tag{47}$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages.

Here the null generators of the horizon are taken to be affinely parametrised as $k^\mu = (\partial/\partial u)^\mu$ in double-null coordinates, so that $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$ is the total null energy of the photon. This differs from the Killing-normalised convention by a factor of the surface gravity $\kappa = a$; the two are related by $(\partial/\partial u)^\mu = a(\partial/\partial\eta)^\mu$ at the horizon, consistent with the standard result $\Delta A = 8\pi G E/\kappa$ when $E = \hbar\omega$ is the Killing energy.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each AS pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [5] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [4] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but his program identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process. The shockwave S-matrix program of 't Hooft [42] established that the longitudinal AS shift is the fundamental dynamical ingredient for horizon information exchange, working at the level of the full S-matrix; the present construction realises the same shift at the level of a single microstate. Each measurement event produces one AS pp-wave, which deposits one supertranslation at the Rindler horizon, exciting one mode of the BMS_3 algebra. This microstate structure is consistent with Carlip's result: the microstates being counted are the measurement events themselves.

This places the bilocal construction inside the soft-theorem / gravitational-memory / BMS triangle established by Strominger and Zhiboedov [6]. Each measurement event sits at the intersection of three equivalent descriptions: a gravitational memory burst (the AS longitudinal shift $\Delta v \propto -\ln r^2$), a zero-energy soft graviton insertion (a pole in the Weinberg soft graviton theorem [43]), and a BMS supertranslation mode on the horizon. The bilocal source makes this triangle constructive rather than merely structural: one emitted photon pair produces one AS pp-wave, one supertranslation, and one excited mode of the BMS_3 algebra, in a one-to-one correspondence. The counting of horizon entropy via the BMS_3 central charge [4] then acquires a direct operational reading: it counts measurement events, one supertranslation mode per absorbed photon.

5.5 The Localization Transition

The planar impulsive wave and the AS pp-wave are the two extremes of a one-parameter family parametrised by the transverse width σ of the photon mode. Replacing the uniform density $\hbar\omega/A_\perp$ in (37) with a normalized Gaussian $g_\sigma(\mathbf{x}_\perp) = (\pi\sigma^2)^{-1}e^{-r^2/\sigma^2}$,

$$T_{uu}(u, \mathbf{x}_\perp) = \hbar\omega\, g_\sigma(\mathbf{x}_\perp)\, \delta(u), \tag{48}$$

gives a family that connects the two limits: as $\sigma \to \infty$ (at fixed horizon area $A_\perp$) the energy density becomes approximately uniform and the shift approaches the quadratic form (42); as $\sigma \to 0$ the energy concentrates on the axis and the shift approaches the AS logarithmic form (93). The total null energy $\hbar\omega$ is preserved throughout, so

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp = 8\pi G\hbar\omega \tag{49}$$

holds for all σ. The regulated longitudinal shift of Appendix 11 describes this interpolation in the neighbourhood of the AS limit; the natural transverse scale is the photon wavelength $\sigma \sim c/\omega$, consistent with the regularization scheme of that appendix.

The BMS mode content transitions continuously with σ. At large σ the Laplacian $\nabla_\perp^2(\Delta v)$ is approximately constant, selecting the $\ell = 2$, $m = 0$ mode; as σ decreases, the Laplacian sharpens toward a delta function and progressively higher multipoles are populated until the full BMS spectrum is excited in the AS limit.

The localization transition is therefore simultaneously a transition in the BMS mode content of the horizon perturbation, a transition in the spatial distribution of the area increment, and the geometric counterpart of the POVM-to-PVM transition of Section 3.2: the one-parameter Kraus family (14) parametrized by σ has a direct geometric image in the interpolation between the planar impulsive wave and the AS pp-wave, with the Stokes line of [20] marking the boundary between the two regimes.

5.6 Action and the Gravitational Boundary

The equality derived below is an internal consistency check on the microstate identification, not a novel result claimed in isolation. The thermodynamic interpretation of the GHY boundary term on null surfaces is established by Chakraborty and Padmanabhan [44]; the broader program of Padmanabhan [45] identifies gravitational boundary terms as the fundamental degrees of freedom at the ensemble level. What the present construction contributes is not the identity $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ but its realisation microstate by microstate: if individual quantum measurement events genuinely are the microstates underlying each area increment, the geometric count and the informational count must agree event by event. The calculation below verifies that they do, closing the consistency loop.

Each absorption event transfers null energy $\hbar\omega_i$ across the horizon. The matter action at the horizon is defined as the null-surface integral of the stress-energy, evaluated over the horizon generators:

$$S_{\text{matter}} \;=\; \sum_i \int T^{(i)}_{uu}\, du\, d^2x_\perp \;=\; \sum_i \hbar\omega_i, \tag{50}$$

where the second equality follows from energy conservation applied to each photon. This is the null-surface analog of the worldline action $\int E\, d\tau$: on a null surface the affine parameter u plays the role that proper time plays on a timelike worldline, and the integral $\int T_{uu}\, du$ plays the role of $\int E\, d\tau$.

The two are not the same integral. After the absorption event, the detector's worldline action continues to accumulate as $\hbar\omega_i\, \Delta\tau$ for all subsequent proper time $\Delta\tau$, encoding the Bremermann memory of the event. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ holds for the null-surface version of the matter action, evaluated at the moment of horizon crossing; it is a statement about the handoff event itself, not about the subsequent worldline evolution. The Rindler proper time η and the affine null parameter u are related by $u \propto e^{-a\eta}$, and it is this exponential relation, not a simple proportionality, that is responsible for the Unruh temperature and that distinguishes the null-surface action from the worldline action. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (47), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{51}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [46] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{c^4}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{c^4}{8\pi G}\sum_i \Delta A_i = \frac{c^4}{8\pi G}\cdot 8\pi G \sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{52}$$

Chakraborty and Padmanabhan [44] showed that the gravitational boundary term on null surfaces has a thermodynamic interpretation as a heat density Ts of the null surface, establishing $S_{GHY}|_{\mathcal{H}}$ = heat content at the level of thermodynamic averages. The present result realises the same identity at the level of individual microstates, with each contribution to S_{GHY} sourced by a specific quantum measurement event rather than a bulk thermal average.

The matter action and the gravitational boundary action are therefore equal, event by event, on the stationary Rindler horizon. This is not a bulk result: the bulk Einstein–Hilbert action vanishes on-shell, and it is the boundary term that carries the physical content. The microstate construction identifies what that boundary content is: each contribution to S_{GHY} corresponds to one quantum of action deposited into matter-memory by one measurement event.

The equality $S_{GHY}\big|_{\mathcal{H}} = S_{\text{matter}}$ invites a stronger reading. The bulk Einstein–Hilbert action vanishes on-shell, so it is the boundary term alone that carries the gravitational degrees of freedom. If the boundary term counts matter action quanta event by event, as equation (52) shows, then the geometry at the horizon is not an independent quantity that happens to equal the matter action: it *is* the matter action, expressed in geometric language. The Planck area $\ell_P^2 = \hbar G/c^3$ is the conversion factor between the two descriptions, relating one quantum of action $\hbar$ to one Planck-area patch on the horizon. In this reading the metric near the horizon is not a smooth field that matter happens to perturb; it is a coarse-grained count of the action quanta that have passed through the horizon surface, with the Bekenstein–Hawking area law as the statement that these two counts agree. Whether this identification extends from the boundary into the bulk, that is, whether the Ricci scalar R can be understood as an action-quanta density with ℓ_P^2 as the conversion factor, is a question the present construction motivates but does not settle, and we leave it for future work.

This result is consistent with and extends programs of Jacobson [1] and Padmanabhan [45], which argue that the gravitational boundary term rather than the bulk Einstein–Hilbert action is the fundamental gravitational degree of freedom. Those programs establish this at the level of thermodynamic averages. The present construction realizes the same identification at the level of individual microstates, with each boundary contribution sourced by a specific quantum measurement event rather than a bulk average. The standard result emerges from summing over the ensemble; the microstate picture supplies what the thermodynamic programs leave open: the physical process underlying each boundary contribution.

The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is structurally related to the Ryu–Takayanagi formula [47], which identifies the entanglement entropy of a boundary region with the area of a minimal bulk surface in AdS: $S_{\text{ent}} = A/4G\hbar$. Three differences sharpen the comparison. First, the present result holds in flat Minkowski space on a Rindler horizon rather than on a minimal surface in AdS, and requires no holographic dual. Second, it is an equality of actions rather than of entropies, with the Planck area ℓ_P^2 as the conversion factor between the two sides rather than as a fixed coefficient. Third, it holds event by event rather than as a thermodynamic average over a thermal ensemble: the AS pp-wave sourced by a single photon produces a single calculable area increment, and summing over events recovers the thermodynamic result. Whether $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is the flat-space Rindler limit of the Ryu–Takayanagi formula, or whether the two results are independent derivations of the same underlying holographic principle from orthogonal directions, is a natural question for future work.

The classical focusing theorem used above to derive $\Delta A = 8\pi G\hbar\omega$ has a quantum generalisation, the Quantum Null Energy Condition (QNEC) [48, 49], which replaces the null energy condition with a bound involving the second derivative of von Neumann entropy of the outgoing field state along the null generators. The QNEC is the natural framework for extending the present construction beyond pure microstates: the classical focusing theorem suffices here because the bilocal construction maintains a globally pure state throughout, with no need to trace out either wedge. At the boundary of EFT validity, where the single-wedge Rindler description requires tracing out $\mathcal{L}$ and the global pure state gives way to a mixed thermal state, the QNEC becomes the operative inequality; that it saturates precisely at $\omega = \omega_{\text{sat}}$ is consistent with the six equivalent conditions of Proposition 1, and we leave its precise role at that boundary for future work.

6 Entanglement as Spatial Information

The two geometric regimes of Section 5 carry complementary entanglement interpretations. In the delocalized limit, the planar impulsive wave uniformly displaces the entire horizon outward in the $+v$ direction (right wedge) and $-v$ direction (left wedge): the two wedges are pushed apart globally, and the geometric connection between them is generated by the entanglement of the full thermofield double state (24). This is the ensemble-level version of Van Raamsdonk's argument [3]: the single BMS mode excited by the delocalized photon corresponds to a global deformation of the geometry connecting the two wedges. In the localized limit, the AS longitudinal shift is concentrated in a cylindrical neighbourhood of the photon trajectory: the two wedges are displaced apart locally, microstate by microstate, and the geometric connection is built up event by event from individual measurement interactions. The full Van Raamsdonk picture is the sum of these local events over the ensemble; the bilocal construction makes each contributing event explicit.

Van Raamsdonk [3] showed, in the AdS/CFT setting, that the entanglement entropy between two halves of a thermofield double state controls the geometry connecting them: reducing entanglement elongates the ER bridge until the geometry disconnects entirely. The vacuum decomposition (24) is precisely the thermofield double state [50], now in flat Minkowski space with helicity labels carried through. The AS longitudinal shifts of Section 5 are its event-by-event geometric output: each measurement event displaces the two wedges apart in opposite null directions, creating new coordinate space between them.

A potential sign puzzle deserves immediate attention. Van Raamsdonk's argument implies that a maximally entangled state (such as the Bell state (26)) should produce maximal geometric *connection* between the two sides, whereas the AS shifts push the wedges *apart*. The resolution is that outward displacement in the present construction is not disconnection but creation of interior volume: the two wedges are not separating topologically but making room for the geometry between them, consistent with the positive longitudinal shift generating new Minkowski space rather than a change in causal structure. This is the event-by-event realisation of Van Raamsdonk's entanglement-generated geometry. Three features go beyond his argument: the setting is flat space without an AdS/CFT dual; the result holds microstate by microstate rather than at the level of entanglement entropy; and the source $\mathcal{P}$ is explicit, providing the physical mechanism that generates the entanglement.

The bilocal source prepares an anticorrelated Bell state (26). We consider the simplest case: both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and both obtain a definite outcome, which we label H_+ or H_-. In this case their outcomes are perfectly anticorrelated by the Bell state structure, independently of which basis is chosen. The question

"who observed H_+?"

then has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*, a bit that encodes not a polarization state but a location in space. The spatial bit is where the observer-relativity of the construction becomes acute. From $\mathcal{R}$'s perspective the answer to *"who observed H_+?"* is a definite classical fact, recorded in matter-memory. From $\mathcal{P}$'s perspective before any measurement, the answer is indeterminate: the Bell state assigns no definite location to H_+. From the full Minkowski perspective after both measurements, the two records are perfectly correlated but the question of *when* the spatial fact became definite has no observer-independent answer. A PVM by $\mathcal{R}$ on a Bell state does not reveal a pre-existing spatial fact; it *creates* the spatial bit as a classical record in $\mathcal{R}$'s matter-memory. This is the same PVM projection as Section 3.2, now read informationally rather than geometrically: the projection that selects the localized microstate from the thermofield double is the same event as the projection that creates the spatial bit in matter-memory.

Whether that creation is an objective event or a relational one is a question the present construction does not need to resolve in full generality. Within this construction, the spatial bit is relational but not arbitrary: its creation is grounded in the common causal ancestry of $\mathcal{P}$, which enforces the anticorrelation between $\mathcal{R}$'s and $\mathcal{L}$'s records as a consequence of shared causal history rather than an objective collapse.

Each measurement event has exactly two irreducible constituents: one quantum of action $\hbar$ written persistently into the detector's matter-memory as a series of samples, and one bit of polarization registered as a classical outcome. The entanglement between $\mathcal{R}$ and $\mathcal{L}$ is consumed in producing these two records, an increase in sampling rate and one bit at each wedge. What remains after measurement is not the entanglement itself but its classical residue: the perfect anticorrelation between the two bits, and the correlated pair of samples written into the matter on each side of the horizon. The spatial bit is precisely the polarization bit now read as a location rather than a spin state: the same bit that records *what* was observed also records *where*, because in this construction the two wedges carry opposite helicities by design. This double duty is forced: the anticorrelated helicity assignment of the bilocal source makes the two wedges distinguishable by construction, so the spatial bit is already latent in the source geometry before any measurement occurs.

The information is carried by the matter of the detectors, not by the horizon. As established in Section 4, the horizon is a boundary that makes the left/right distinction sharp; it is the matter on each side that accumulates the samples and holds the classical record. $\mathcal{R}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_+/H_-. $\mathcal{L}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_-/H_+. The two records are perfectly complementary by the anticorrelated structure of the bilocal source, and their complementarity is encoded in matter and is reflected in geometry.

6.1 Emergent Space and the Thermofield Double

The Unruh vacuum is the thermofield double state of the left and right Rindler wedges [51]: equation (24) is precisely this state, with the helicity label λ carried through. The bilocal source selects one term from this sum at fixed ω and fixed helicity pair, yielding the single microstate (26). The correlated AS longitudinal shifts in opposite null directions are the geometric output of this microstate: a linearized perturbation of flat Minkowski space that displaces the two wedges away from their respective horizons, creating new coordinate space between them.

The connection between the two wedges requires no exotic geometry: $\mathcal{R}$ and $\mathcal{L}$ are correlated because they share a common causal past in $\mathcal{P}$, which is ordinary retarded causality. Non-traversability is equally automatic, no signal can pass directly from $\mathcal{L}$ to $\mathcal{R}$ because the causal structure of Minkowski space forbids it, matching the no-communication theorem on the EPR side without additional assumptions. The ER=EPR correspondence, in this flat-space setting, reduces to the statement that entangled systems with a common causal ancestor generate correlated geometric displacements, a consequence of energy–momentum conservation at $\mathcal{P}$ rather than a postulate about wormhole topology. The wormhole picture of ER=EPR may be what this flat-space causal structure looks like in AdS, when the source $\mathcal{P}$ is not explicitly available and the connection between wedges must be encoded in the geometry itself.

6.2 Observer-Relativity and the Multi-Observer Problem

The construction is naturally multi-observer rather than single-frame. Standard QFT in the S-matrix formulation assumes a single asymptotic observer who prepares the initial state and measures the final state; in the language of Section 4.6, there is one lab and one measurement at the end. Horizon physics requires a different framework: four observers $\mathcal{P}$, $\mathcal{R}$, $\mathcal{L}$, $\mathcal{F}$ each hold partial, causally complementary information about the same physical process. Neither $\mathcal{R}$ nor $\mathcal{L}$ alone can reconstruct the emission at $\mathcal{P}$; their two classical bits together can. The horizon is not a barrier in a single-observer description but a surface that makes the multi-observer structure of the construction sharp: it is the boundary at which $\mathcal{R}$'s and $\mathcal{L}$'s causal domains separate. This is not a limitation of the framework but its natural setting, and it is structurally the same situation that arises in Wigner's friend scenarios [36] applied to gravitational systems, a point developed in the following section.

The multi-observer structure exposed here is not merely an analogy with Wigner's friend: it is precisely that scenario, instantiated in a gravitational context with causal separation enforced by a horizon rather than by a laboratory door.

A complete formalism for multi-observer quantum mechanics, one that specifies consistently how causally separated observers' measurement records combine into a single description, does not yet exist. The Wigner's friend no-go theorems [36] and the Frauchiger–Renner result [37] establish that the problem is sharp; the relational, QBist, and Everettian programs each dissolve it differently rather than solving it. The present construction operates within a single observer's description but does not treat the gap as a liability: the observer-independence of the Bekenstein–Hawking area law emerges here not from a completed multi-observer formalism but from the common causal ancestry of $\mathcal{P}$, which enforces intersubjective consistency between $\mathcal{R}$'s and $\mathcal{L}$'s records without requiring an objective wavefunction. The Frauchiger–Renner and Wigner's friend problems remain open in general; the construction shows that causal ancestry is sufficient to ground intersubjective agreement in this specific setting.

Whether the relational [38], QBist [39], or Everettian frameworks are inequivalent in this setting is an open question with physical consequences: different resolutions imply different answers to whether $\mathcal{R}$'s microstate count and $\mathcal{L}$'s microstate count are consistent as objective facts or only as relational ones. The information-theoretic objects introduced here, the spatial bit, the PVM projection, the handoff event, the matter-memory, are the vocabulary in which that question must eventually be posed, but they do not answer it. Multi-observer quantum mechanics in the presence of horizons is largely uncharted, and the present construction should be read partly as a map of that problem space.

7 Carnot Cycle from Unruh–Landauer

The Carnot cycle developed here is evaluated at the saturation frequency $\omega_{\text{sat}} = a \ln 2/c$ of Section 4.4, the unique frequency at which a single photon deposits exactly one bit of Bremermann channel capacity per Rindler period. At this frequency the Unruh temperature $k_B T_H = \hbar a/2\pi c$ and the photon energy $\hbar\omega_{\text{sat}} = \hbar a \ln 2/c$ are related by

$$k_B T_H = \frac{\hbar\omega_{\text{sat}}}{\ln 2}, \tag{53}$$

which is $k_B T_H \sim \hbar\omega_{\text{sat}}$ up to the factor $1/\ln 2 \approx 1.44$, the nats-to-bits conversion. This factor has a precise meaning in the present framework: it is the ratio between the natural information unit of the Unruh thermal distribution (nats, set by e) and the binary unit of the helicity outcome (bits, set by $\ln 2$). Throughout this section we absorb this factor into the order-of-magnitude identification

$$k_B T_H \sim \hbar\omega_{\text{sat}}, \tag{54}$$

consistent with the heuristic character of the Schwarzschild radius and surface gravity derivations of Section 2, where numerical prefactors of the same order appear for the same reason. The precise coefficient is recovered by working in nats throughout, in which case $\ln 2 \to 1$ and equation (53) becomes exact. The six equivalent conditions of the upcoming Proposition 1 all reduce to $\omega = \omega_{\text{sat}}$ at the horizon, so evaluating the cycle at this frequency is not an additional assumption but the condition that the EFT, which we will explore in the next section, is at its natural boundary.

The Carnot cycle developed here is formulated at the level of individual non-thermal microstates, and this is the point. Reversibility is a feature of the microstate description: in a coarse-grained thermal treatment individual microstates are inaccessible and the cycle could not be run explicitly. Temperature appears in this section not as a primitive imported from the Rindler description but as a derived label on each leg of the cycle, consistent with the non-thermal Minkowski vantage point maintained throughout. The Carnot efficiency bound and the Unruh–Landauer bound will emerge as the same inequality viewed from opposite sides of this microstate-level construction, closing the loop opened in Section 2.

We consider in this section the case where $\mathcal{L}$ and $\mathcal{R}$ are moving mirrors being accelerated with an outward thrust from our past bilocal source $\mathcal{P}$ emission, see Figure 7. We complete our bilocal microstate absorption example by reflecting and emitting the photons off of the mirrors.

This reflects the energy back to the future sink $\mathcal{F}$, located at $(t, z) = (1, 0)$, which is the PCT image of $\mathcal{P}$ under the combined time-reversal and parity operation that swaps past and future wedges. $\mathcal{F}$ plays the role of a final absorber, receiving the reflected photons and resetting the field state; its existence is required by the reversibility of the ideal mirror cycle and by the PCT symmetry of the bilocal construction. The reflection is reversible in the sense that the mirrors do not retain a persistent copy of the information that they temporarily store.

The mirrors are ideal, in the sense that they do not retain a persistent copy of the information that they temporarily store; they forget and emit. It is this reversibility that allows us to construct a example of a Carnot cycle at the level of microstates. The working system is the photon–mirror interaction degree of freedom; the Rindler wedge defines the thermal reservoirs, and the mirror trajectory defines the work parameter.

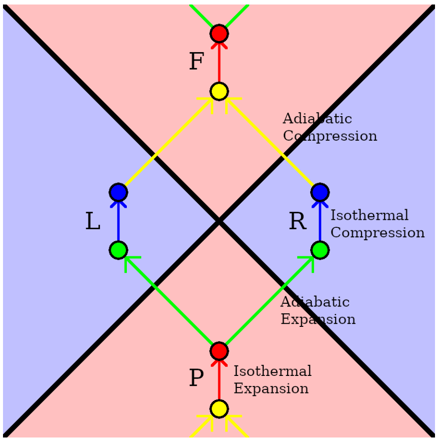

Figure 7: Carnot Cycle: Past ($\mathcal{P}$) bilocal source emits entangled pair with one leg hitting a moving mirror in the right wedge ($\mathcal{R}$). Reflection focuses to a future sink ($\mathcal{F}$). A similar balancing situation exists in the left wedge.

Consider the point of view of the right Rindler observer $\mathcal{R}$ who sees the event horizon as a thermal object. The Rindler horizon plays the role of the hot reservoir at the Unruh temperature $T_H = \hbar a/2\pi k_B c$, consistent with the standard identification of the horizon as a thermal object and its connection to the generalised second law [52]. The Rindler wedge interior, where the mirror operates, is taken as the cold reservoir at temperature T_C. The identification of T_C with the Landauer erasure temperature of the mirror is an ansatz of the present construction; its precise value determines the Carnot efficiency $\eta = 1 - T_C/T_H$.

- **Isothermal Expansion:** The bilocal source prepares an entangled photon pair in the basis that $\mathcal{R}$ measures in (H_+/H_- see section 6). In this basis, the Shannon entropy of the Rindler outcome distribution increases from 0 to 1 bit, reflecting the transition from a deterministic prior state to a maximally uncertain Bell state induced by the bilocal entanglement structure, , i.e. the inverse of the PVM projection of Section 3.2: the isothermal expansion runs the projection backward, from a sharp microstate to the full superposition; notionally "drawing some heat from the environment $\mathcal{P}$".

- **Adiabatic Expansion:** This corresponds to the crossing over the past Rindler horizon from hot to cold, the absorption of the photon by the mirror.
- **Isothermal Compression:** This corresponds to controlled erasure of the detector's stored bit during the re-emission of the photon by the mirror. This is where the Landauer cost is paid [11, 53]: Bennett showed that measurement can in principle be performed reversibly; it is the erasure of the stored bit at re-emission that necessarily dissipates energy, at minimum $k_B T_C \ln 2$ into the cold reservoir.
- **Adiabatic Compression:** This corresponds to the crossing over the future Rindler horizon from cold to hot, the absorption of the future sink $\mathcal{F}$ and reseting of the state to an eigenstate of H_+ or H_-, zero bits of Shannon entropy.

7.1 Shannon Entropy, QBism, and Wigner's Friend

The entropy variation in the isothermal legs should be understood as Shannon entropy associated with measurement outcomes in a fixed operational basis, rather than the von Neumann entropy of the global quantum state, which remains pure throughout. This distinction has a precise foundation. The four-observer structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$–$\mathcal{F}$ is a gravitational realisation of the Wigner's friend scenario [36]: $\mathcal{P}$ prepares the Bell state (Wigner's role), $\mathcal{R}$ and $\mathcal{L}$ each measure it (the friends' roles), and the question of when "collapse" occurred is observer-relative. From $\mathcal{R}$'s perspective, the state collapsed upon photon absorption; from the full Minkowski perspective, the global state remained pure throughout. These descriptions are not contradictory but complementary.

The QBist framework [39] resolves this cleanly, and is the natural home of the present construction. In QBism, quantum states encode an agent's beliefs rather than objective microstate facts; the correlations between agents' outcomes are objectively real and correctly predicted by the Bell state structure, but no single agent holds the "true" global state. Each observer's Shannon entropy, the uncertainty in their helicity outcome prior to measurement, is valid relative to them. The Carnot cycle is therefore formulated relative to $\mathcal{R}$: its Shannon entropy increases from zero to one bit on the isothermal expansion leg, decreases on the isothermal compression leg, and the area of the cycle in T-H space is the work extracted, all defined by $\mathcal{R}$'s operational measurement basis.

No decoherence assumption is required; Shannon entropy here is defined over the classical outcome distribution induced by a fixed measurement basis (the Rindler detector algebra), independent of the unitary evolution of the global state. The objectivity of the Bekenstein–Hawking area count is not in tension with the observer-relativity of the individual microstate: the area increment $\Delta A = 8\pi G\hbar\omega$ is a covariant physical event, while the assignment of that event to a particular observer's measurement record is relational. Thermodynamic entropy, in this reading, is the classical face of quantum observer-relativity: it measures what a given agent does not yet know, and its increase under measurement is the same process as the Bayesian update arrow of Figure 1, now with a thermodynamic label attached.

The interpretational position underlying the construction deserves to be stated explicitly. Quantum information is treated here as an extension of classical information rather than a replacement for it. A Hilbert space of dimension 2^n for n qubits has the same dimensional structure as the probability space of n classical bits: the statement that two coins are correlated as HH or TT but never HT or TH already requires a 2^2-dimensional description, and the Bell state (26) is the quantum analog of exactly such a classical correlation, with probability amplitudes replacing probabilities. Quantum observation is correspondingly the quantum analog of Bayesian conditionalization: the PVM projection updates an agent's probability assignment upon receipt of a classical outcome. No commitment to a specific ontology is required to say this.

The stand the construction takes enters precisely at the PVM rule: each handoff event is a projective measurement, and its outcome, the helicity eigenvalue, the spatial bit, is recorded in the observer's matter-memory as a persistent classical fact. Memory is what grounds an observer in their particular sequence of outcomes. To interfere with an observer who recorded H_+ would require erasing that record; without erasure, the memory distinguishes the two outcomes and no interference is possible. The Carnot cycle's isothermal compression step is the precise physical implementation of this: the mirror forgets the bit it learned, paying the Landauer cost $k_B T_C \ln 2$. The ideal mirror limit $T_C \to 0$ is the limit of perfect forgetting at zero thermodynamic cost; the condition $T_C > 0$ is the condition that some memory persists and some coherence is permanently lost. The paper does not resolve whether perfect forgetting constitutes a transition between Everettian branches or a local reset of the field state; it shows that the thermodynamic cost of forgetting is real and precisely calculable, and that the spatial bit, the PVM, the matter-memory, and the Landauer erasure are the vocabulary in which that question must eventually be posed.

7.2 Work and Efficiency

The thermodynamic state space of the cycle is parameterised by temperature T and Shannon entropy H. The two isothermal legs operate at T_H and T_C respectively, and the entropy change per cycle is exactly one bit:

$$\Delta H = \ln 2 \quad \text{(one binary helicity outcome)}, \tag{55}$$

so the work extracted per cycle is the area of the rectangle in T-H space:

$$W = (T_H - T_C)\,\Delta H = (T_H - T_C)\ln 2. \tag{56}$$

The work done on the mirror by absorbing one photon of energy $\hbar\omega$ is the radiation pressure impulse $\Delta p = \hbar\omega/c$, which is responsible for the mirror's acceleration into the Rindler wedge. This identifies the hot reservoir temperature via

$$k_B T_H = \hbar\omega, \tag{57}$$

the energy of one photon sets the Unruh temperature of the horizon. The Carnot efficiency bound $W \le T_H \Delta H$ then gives

$$W \le \hbar\omega \ln 2 = \frac{\hbar a \ln 2}{2\pi c}, \tag{58}$$

where in the last step we used $k_B T_H = \hbar a / 2\pi k_B c$ from the Unruh temperature. This is precisely the Unruh–Landauer bound (3) at $\Delta H = \ln 2$:

$$\boxed{W \le \frac{\hbar a}{2\pi c}\Delta H \iff \Delta E \ge \frac{\hbar a}{2\pi c}\Delta H.} \tag{59}$$

The Carnot efficiency bound for this cycle and the Unruh–Landauer bound are therefore the same inequality, approached from opposite directions: the Carnot bound is a statement about the maximum work extractable from the thermal gradient between the horizon and the wedge interior, while the Unruh–Landauer bound is a statement about the minimum energy cost of acquiring one bit at that horizon. Their equivalence identifies the Rindler horizon as a Carnot engine operating at the Unruh temperature, with quantum measurement as the working substance and the classical bit as the thermodynamic output.

The classical precursor to this equivalence is the Geroch process [14]: a box of radiation lowered toward a black hole on a string extracts work until, at the horizon, the box carries zero energy from the perspective of an observer at infinity. Bekenstein showed that dropping the box in must increase the black hole's area, establishing a maximum efficiency for work extraction that saves the second law [14]. Our mirror plays the role of the box, the measurement impulse plays the role of the string tension, and the Unruh–Landauer bound is Bekenstein's efficiency constraint expressed in the language of information acquisition at the Rindler horizon. The two bounds are not merely analogous; they are the same inequality, with the Geroch process as its classical limit and the Carnot cycle here as its microstate realisation.

The cold reservoir temperature T_C is the Landauer temperature associated with isolating the mirror from its local re-emission environment, while T_H is the Landauer temperature associated with isolating the entire Rindler wedge from the global Minkowski vacuum. The two temperatures correspond to two levels of coarse-graining: T_H traces out the opposite wedge at the Rindler horizon, while T_C traces out only the mirror's local photon field at the scale of the mirror interaction. The temperature gradient $T_H > T_C$ is therefore a gradient between two scales of isolation, and the Carnot efficiency $\eta = 1 - T_C/T_H$ measures how much of the global thermal gradient is captured by the local mirror interaction. In the limit $T_C \to 0$ the mirror's local environment is fully decoupled and the cycle extracts maximum work; in the limit $T_C \to T_H$ the two scales of isolation coincide and the cycle is in hydrostatic equilibrium with no net work extracted.

The efficiency $\eta = 1 - T_C/T_H$ is maximised in the limit $T_C \to 0$, which corresponds to perfect erasure at zero thermodynamic cost, the ideal mirror limit. Any residual $T_C > 0$ reflects irreversibility in the mirror's erasure process, consistent with Landauer's principle applied to the re-emission step.

8 Effective Field Theory Structure

The bilocal construction has a natural effective field theory interpretation. The right wedge $\mathcal{R}$ is the *system*, with energy scale E_{sys}; the full Minkowski space $\mathcal{R} \cup \mathcal{L}$ is the *environment*, with energy scale $E_{\text{env}} = E_{\text{sys}} + E_{\mathcal{L}}$. The Rindler horizon is the EFT cutoff surface: it is the boundary at which the system's description and the full Minkowski description become incompatible, and the left wedge $\mathcal{L}$ is precisely the degrees of freedom that the system has integrated out.

8.1 The GHY Term as Effective Action

We now show that the GHY boundary term arises as the effective action for $\mathcal{H}_R$ after integrating out $\mathcal{H}_L$ in the path integral, elevating the identification $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ from Section 5 from an algebraic observation to a structural result.

Setup. The full Euclidean path integral for gravity and matter decomposes over the two wedges as

$$Z = \int \mathcal{D}g_R\, \mathcal{D}\phi_R \int \mathcal{D}g_L\, \mathcal{D}\phi_L\; e^{-S_R - S_L - S_{\mathcal{H}}}, \tag{60}$$

where $S_{\mathcal{H}}$ is the boundary term at the horizon coupling the two wedges. Integrating out the left wedge defines the effective action:

$$e^{-S_{\text{eff}}[g_R,\phi_R]} = \int \mathcal{D}g_L\, \mathcal{D}\phi_L\; e^{-S_L - S_{\mathcal{H}}}. \tag{61}$$

Origin of the horizon coupling. Integration by parts applied to any kinetic term produces a boundary contribution at $\mathcal{H}$. For the gravitational field this is the Gibbons–Hawking–York term [46]

$$S_{\mathcal{H}} = S_{GHY}\big|_{\mathcal{H}} = \frac{1}{8\pi G} \int_{\mathcal{H}} d^2x_\perp\, d\lambda\, \theta, \tag{62}$$

which couples the two wedge geometries through their expansion θ at the horizon, precisely as a scalar kinetic term couples ϕ_L and ϕ_R through their normal derivatives.

Saddle point evaluation. The classical solution in the left wedge is flat Minkowski space perturbed by the AS pp-wave of the left-moving photon. Aichelburg–Sexl pp-waves are Ricci flat ($R_{\mu\nu} = 0$) in vacuum, so the bulk Einstein–Hilbert action vanishes on-shell:

$$S_L\big|_{\text{on-shell}} = \frac{1}{16\pi G}\int_L d^4x\sqrt{g}\,R = 0. \tag{63}$$

In a standard EFT path integral, integrating out the left wedge would produce two contributions: the gravitational boundary term S_{GHY} (from integration by parts of the bulk kinetic term) and the matter action S^L_{matter} (from the left-moving photon). Treated as independent, these would give $e^{-S_{GHY}-S^L_{\text{matter}}}$, and the effective action would be $S_R + S_{GHY} + S^L_{\text{matter}} = S_R + 2\hbar\omega$.

The central result of Section 5 resolves this: S_{GHY} and S^L_{matter} are not independent contributions but two descriptions of the same physical event, one photon crossing the horizon, counted geometrically and informationally. Adding them separately double-counts that event. The correct horizon coupling is therefore $S_{\mathcal{H}} = S_{GHY}$ alone, with S^L_{matter} already subsumed into the geometric boundary term. At the saddle point:

$$\int \mathcal{D}g_L\,\mathcal{D}\phi_L\, e^{-S_L - S_{\mathcal{H}}}\bigg|_{\text{saddle}} = e^{-S_{GHY}|_{\mathcal{H}}}, \tag{64}$$

where $S_L|_{\text{on-shell}} = 0$ by Ricci-flatness, and the effective action for the right wedge is

Since $S_{GHY}|_{\mathcal{H}} = S^L_{\text{matter}} = \sum_i \hbar\omega_i$ by equation (52), the effective action for the right wedge is

$$\boxed{S_{\text{eff}}[g_R, \phi_R] = S_R[g_R, \phi_R] + S_{GHY}\big|_{\mathcal{H}}.} \tag{65}$$

The GHY term is the contribution to the right wedge effective action from integrating out the left wedge at the classical level. The bulk left wedge action vanishes by Ricci-flatness of the AS solution; the entire left wedge effect is encoded in the boundary term at the horizon.

Quantum corrections. Fluctuations of ϕ_L around the saddle generate corrections to S_{eff} suppressed by ℓ_P^2/A relative to the classical term. These are the standard entanglement entropy contributions studied by Susskind–Uglum [54] and Callan–Wilczek [55], whose UV divergences renormalize Newton's constant G. Equation (65) is the leading classical contribution.

8.2 Partition Function and S-Matrix

The thermal partition function of the right wedge

$$Z = \mathrm{Tr}_{\mathcal{H}_R}\, e^{-H_R/k_B T_H} \tag{66}$$

is obtained from the full Minkowski vacuum by tracing out $\mathcal{H}_L$: integrating out the left wedge in the path integral yields the thermal density matrix for $\mathcal{H}_R$, and $\ln Z$ generates the connected correlation functions of the right wedge EFT. The S-matrix of the full bilocal construction connects the past state at $\mathcal{P}$ to the future sink $\mathcal{F}$:

$$S = \langle \mathcal{F}|\mathcal{P}\rangle = \mathrm{Tr}\, e^{-iHt/\hbar}, \tag{67}$$

which under Wick rotation $\tau = it$ becomes $Z = \mathrm{Tr}\, e^{-H\tau/\hbar}$. The Carnot cycle of Section 7 is therefore the thermodynamic avatar of this S-matrix element: the real-time bilocal construction and its thermodynamic description are related by Wick rotation, and each microstate (26) contributes one term to Z rather than the full thermal trace.

8.3 Horizon Complementarity

The identification of $\mathcal{H}_R$ as system and $\mathcal{H}_L \otimes \mathcal{H}_R$ as environment is the operational content of horizon complementarity [56]: the right wedge observer sees a thermal state at T_H, while the full Minkowski observer sees a pure entangled state. These are the EFT and UV descriptions of the same microstate. The spatial bit, the answer to "who observed H_+?", is visible to $\mathcal{R}$ as a classical record but its quantum origin in the entangled pair is accessible only to the full Minkowski observer who holds both wedges. Complementarity is not an additional postulate here but an operational consequence of the EFT structure: the horizon is the surface at which the single-wedge and two-wedge descriptions become incompatible, consistently with the equivalence list of the following subsection.

8.4 EFT Validity and the Planck Scale

The construction defines two natural length scales. The system scale is the Compton wavelength of the detector

$$\lambda_C = \frac{\hbar}{m_{\mathcal{R}} c}, \tag{68}$$

the scale below which the single-particle description of $\mathcal{R}$ breaks down. The environment scale is the proper horizon distance

$$d_{\mathcal{H}} = \frac{c^2}{a}, \tag{69}$$

the scale at which the single-bit description saturates. The EFT is valid when these scales are separated, $\lambda_C \ll d_{\mathcal{H}}$; it breaks down when they coincide. The Planck length

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} = \sqrt{\lambda_C \cdot r_s} \tag{70}$$

is the geometric mean of the Compton wavelength and the Schwarzschild radius $r_s = 2Gm/c^2$: it is the unique length scale built from both $\hbar$ and G and is the scale at which $\lambda_C = r_s$, i.e. where the system scale and the gravitational scale coincide. The Planck scale is therefore the fixed point of the EFT: the unique scale at which the quantum and gravitational descriptions become equally valid and the EFT has no regime of applicability.

The Carnot efficiency of Section 7 measures the separation between these scales. With $T_H = \hbar a / 2\pi k_B c$ and $T_C = \hbar\kappa / 2\pi k_B c$, the efficiency

$$\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{\kappa}{a} = \frac{m_{\mathcal{L}}}{m_{\mathcal{R}} + m_{\mathcal{L}}} \tag{71}$$

is the fraction of the total mass residing in the environment, the degrees of freedom integrated out. The following conditions are simultaneously equivalent:

Proposition 1 (EFT Validity). *The following are equivalent for the bilocal construction:*

1. $\kappa < a$ *(mirror surface gravity less than global Rindler acceleration)*
2. $E_{\text{sys}} < E_{\text{env}}$ *(system energy less than environment)*
3. $\eta > 0$ *(Carnot cycle extracts positive work)*
4. $\lambda_C < d_{\mathcal{H}}$ *(Compton scale inside horizon distance)*
5. $\ell_P/\lambda_C \ll 1$ *(equivalently $m_{\mathcal{R}} \gg m_P$, the detector mass is super-Planckian)*
6. $\mathcal{R}$ *is a proper subsystem of its environment.*

The boundary case where all inequalities become equalities is simultaneously the horizon condition, the Planck condition $\lambda_C = r_s = \ell_P$, and the thermodynamic equilibrium condition $\eta = 0$. The EFT breaks down exactly at the horizon, which is not a failure of the framework but its natural boundary condition.

9 Conclusion

We have proposed quantum measurement events as explicit models of gravitational microstates in the Rindler setting, partially answering the question left open by Jacobson and Verlinde of what physical process underlies each individual entropy increment. The construction converges independently with three established programs. Jacobson's thermodynamic derivation of the Einstein equations [1] requires a physical process underlying each local entropy increment; the bilocal measurement event is that process, identified explicitly rather than postulated. Van Raamsdonk's entanglement-generates-geometry argument [3], formulated at the level of reduced density matrices in AdS/CFT, is realised here in flat space microstate by microstate, with the AS longitudinal shift as the concrete geometric output of each entanglement event. The BMS soft-theorem-memory triangle [5, 6] identifies the symmetry structure at the horizon; the bilocal construction populates that structure with a concrete physical process, one measurement event per excited supertranslation mode.

The central result is the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$: the gravitational boundary action and the matter action are the same quantity counted in two languages, with the Planck area ℓ_P^2 as the conversion factor. This follows from a single temperature-eliminated Unruh–Landauer bound, from which standard black hole thermodynamics emerges via Newton's laws, and is consistent across thermodynamic, geometric, information-theoretic, and effective field theory descriptions of the horizon. Matter and horizon are orthogonal information channels: matter the timelike substrate of emergent proper time, the horizon the spacelike surface through which PVM projections onto null geodesics generate emergent space.

A related but distinct novelty concerns the epistemic status of the horizon and its entropy. In the standard treatment, Unruh thermality is observer-dependent but objective within each observer's description: the partial trace over the inaccessible wedge produces a genuinely mixed state, not a state of ignorance about an underlying pure microstate. The present construction inverts this. The global Minkowski state is pure throughout; the thermal state that $\mathcal{R}$ sees is a consequence of two coarse-grainings simultaneously, ignorance of $\mathcal{P}$'s emission event and ignorance of $\mathcal{R}$'s location within the wedge, both of which the PVM projection eliminates explicitly. Thermality is therefore epistemic in a precise sense: it is what remains when the source is unspecified, not an objective feature of the vacuum that geometry imposes on accelerating observers. The horizon plays a corresponding double role. As a causal surface it is ontic, signals cannot pass from $\mathcal{L}$ to $\mathcal{R}$. But as an epistemic boundary it is the surface at which $\mathcal{R}$'s single-wedge EFT description and the global Minkowski description become incompatible, the geometric expression of the point at which ignorance becomes irreducible for a single observer.

Horizon entropy, in this reading, is not a property of the geometry alone but a measure of the epistemic gap between these two levels of description, with the Planck area ℓ_P^2 as the conversion factor between the ontic geometric count and the epistemic informational count. The ontic causal surface and the epistemic entropy surface coincide at the horizon because causal inaccessibility and informational irreducibility are the same condition, approached from the spacetime and information-theoretic sides respectively.

This is what the inversion from $\mathcal{R}$'s passive thermal reading to $\mathcal{P}$'s active emission event achieves, and it is precisely what allows the construction to live in flat Minkowski space: no holographic dual is needed because the geometry is the output of measurement events, not their prerequisite.

The observer-relativity of the microstate construction is a feature rather than a liability. The Bekenstein–Hawking area and horizon entropy are objective, observer-independent quantities; the microstates identified here, PVM projections by a specific Rindler observer $\mathcal{R}$, recording a spatial bit in a specific matter-memory, are observer-relative. These are consistent because observer-relative microstate counts are grounded in the common causal ancestry of $\mathcal{P}$: the anticorrelation between $\mathcal{R}$'s and $\mathcal{L}$'s records is enforced by shared causal history, and the observer-independent area law is the statement that these observer-relative counts are mutually consistent. The remaining open question is not how to resolve the observer-relativity but how to extend this causal grounding to a complete multi-observer formalism, a question the information-theoretic vocabulary introduced here, spatial bits, PVM projections, handoff events, matter-memory, is equipped to pose precisely.

The Frauchiger–Renner theorem [37] shows this is not merely a philosophical gap but a formal one, with measurable consequences; in the gravitational context, with causally separated observers and non-commuting measurement algebras across a horizon, the problem is sharper still. The information-theoretic vocabulary introduced here, spatial bits, PVM projections, handoff events, matter-memory, and the Planck area as conversion factor, is what is needed to state the problem precisely, even if it does not resolve it.

The construction takes no stand on the ontology of the wavefunction. The results, the area increment $\Delta A = 8\pi G\hbar\omega$, the predicted statistics, and the Landauer cost of the erasure step, are the same whether the PVM projection is read as a Bayesian update, a relational fact, or a branching event. The interpretational commitment lies entirely in what one takes the classical record in matter-memory to mean. The relation to the Diósi–Penrose objective collapse program [9, 10] is precise on this point: that program postulates observer-independent gravitational collapse of the wavefunction, whereas the present construction treats state reduction as a Bayesian update for a given observer, with the dashed arrow of Figure 1 the only relation we do not adopt.

Whether the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ extends from the boundary into the bulk, whether the helicity structure of the emergent space constitutes a topological invariant, whether the framework admits falsifiable consequences distinguishing it from standard thermodynamic treatments, and whether the flat-space construction presented here is the Rindler limit of a more general statement about arbitrary horizons, remain open questions for future work.

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10 Appendix A: Consistency of the Double-Null Gauge Condition

We verify that the double-null gauge condition $A_+ = A_- = 0$ is consistent with Maxwell's equations and leaves exactly two physical degrees of freedom.

10.1 Coordinates and Notation

In double-null coordinates (x^+, x^-, x^1, x^2) the Minkowski metric reads

$$ds^2 = -2\, dx^+\, dx^- + (dx^1)^2 + (dx^2)^2, \tag{72}$$

so that $\partial_+ = \partial/\partial x^+$, $\partial_- = \partial/\partial x^-$, and the d'Alembertian is

$$\Box = -2\partial_+\partial_- + \partial_i\partial^i, \qquad i = 1, 2. \tag{73}$$

The gauge field has components $A_\mu = (A_+, A_-, A_1, A_2)$ and transforms as

$$A_\mu \to A_\mu + \partial_\mu\Lambda \tag{74}$$

under a gauge transformation with parameter $\Lambda(x^+, x^-, x^1, x^2)$.

10.2 Imposing the Gauge Conditions in Stages

The single gauge parameter Λ is a function of all four coordinates. We use it in three stages, each consuming a different functional dependence.

Stage 1: $A_+ = 0$. Requiring $A_+ + \partial_+\Lambda = 0$ determines Λ as a function of x^+:

$$\Lambda(x^+, x^-, x^i) = -\int^{x^+} A_+(s, x^-, x^i)\, ds + \Lambda_0(x^-, x^i), \tag{75}$$

where $\Lambda_0(x^-, x^i)$ is an arbitrary residual function independent of x^+.

Stage 2: $A_- = 0$. Under the residual freedom $\Lambda_0(x^-, x^i)$, the A_- component transforms as $A_- \to A_- + \partial_-\Lambda_0$. Setting this to zero determines Λ_0 as a function of x^-:

$$\Lambda_0(x^-, x^i) = -\int^{x^-} A_-(s, x^i)\, ds + \Lambda_1(x^i), \tag{76}$$

where $\Lambda_1(x^i)$ is a residual function of the transverse coordinates only.

Stage 3: Transverse Coulomb condition $\partial^i A_i = 0$. Under the remaining freedom $\Lambda_1(x^i)$, the transverse components transform as $A_i \to A_i + \partial_i\Lambda_1$. Imposing $\partial^i A_i = 0$ requires

$$\nabla_\perp^2 \Lambda_1 = -\partial^i A_i, \tag{77}$$

which is a Poisson equation in the transverse plane with a unique solution (up to harmonic functions, fixed by boundary conditions). This exhausts the gauge freedom entirely.

The three conditions have consumed the full gauge parameter Λ in stages, with no over-determination: each stage uses a strictly different functional dependence of Λ.

10.3 Physical Degrees of Freedom

After imposing $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the remaining components are A_1 and A_2 subject to one differential constraint $\partial^i A_i = 0$. In momentum space this reads $k^i A_i = 0$.

For our specific construction, the photons propagate in the $\pm z$ direction, so their wave-vectors are purely null: $k_\perp^i = 0$ for $i = 1, 2$. The transverse Coulomb condition $k^i A_i = 0$ is therefore trivially satisfied for any A_1, A_2, and both transverse components are independent. The two circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{78}$$

are the two independent physical degrees of freedom, corresponding to helicity $\lambda = \pm 1$. This is the correct count for a massless spin-1 field in four dimensions.

The trivial satisfaction of the transverse Coulomb condition relies on the photons propagating purely in the $\pm z$ direction with vanishing transverse momentum $k_\perp^i = 0$. For photons with nonzero transverse momentum the condition $k^i A_i = 0$ imposes a non-trivial constraint and the degree of freedom count requires revisiting.

10.4 Consistency with Maxwell's Equations

With $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the non-zero field strength components are

$$F_{+i} = \partial_+ A_i, \qquad F_{-i} = \partial_- A_i, \qquad F_{ij} = \partial_i A_j - \partial_j A_i, \qquad F_{+-} = 0. \tag{79}$$

Maxwell's equations $\partial^\mu F_{\mu\nu} = J_\nu$ decompose as follows.

Transverse components $\nu = i$:

$$-2\partial_+\partial_- A_i + \partial^j(\partial_j A_i - \partial_i A_j) = J_i. \tag{80}$$

Using $\partial^i A_i = 0$, the second term simplifies to $\partial^j \partial_j A_i$, giving the standard wave equation

$$\boxed{\Box A_i = J_i.} \tag{81}$$

Null components $\nu = \pm$:

$$-\partial^i \partial_\pm A_i = J_\pm. \tag{82}$$

These are constraint equations relating the null source components to the transverse field. For our bilocal source, which is purely transverse, $J_+ = J_- = 0$, and the constraints reduce to

$$\partial^i \partial_\pm A_i = 0. \tag{83}$$

Differentiating the transverse Coulomb condition $\partial^i A_i = 0$ with respect to $x^\pm$ gives $\partial^i(\partial_\pm A_i) = 0$, so these constraints are automatically satisfied. ✓

10.5 Summary

The double-null gauge condition $A_+ = A_- = 0$ supplemented by the transverse Coulomb condition $\partial^i A_i = 0$:

1. uses the gauge freedom of Λ completely and without over-determination, via a three-stage procedure consuming the x^+-, x^--, and x^i-dependent parts of Λ in sequence;
2. leaves exactly two physical degrees of freedom, the helicity eigenstates $\lambda = \pm 1$, consistent with the known count for a massless spin-1 field;
3. reduces Maxwell's equations to the standard transverse wave equation $\Box A_i = J_i$, with the null constraint equations automatically satisfied for a purely transverse source.

The gauge is therefore internally consistent and does not over-constrain the physical degrees of freedom. The PCT wedge-swap symmetry is preserved throughout: conditions on A_+ and A_- are imposed symmetrically, and neither null direction is privileged over the other.

11 Appendix B: Regularisation of the Aichelburg–Sexl Singularity

The derivation of ΔA in Section 5.4 applies to a normalizable photon mode but inherits from the strict Aichelburg–Sexl construction a distributional stress tensor $T_{uu} \propto \delta^{(2)}(\mathbf{x}_\perp)$ and a longitudinal shift $\Delta v \propto -\ln r^2$ that diverges on the axis $r \to 0$. The electromagnetic construction of Section 3 provides a natural regularisation: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth Gaussian transverse profile of characteristic width $\sigma \sim c/\omega$ set by its wavelength. We carry out the regularisation explicitly here.

Photon stress tensor and energy conservation. In double-null gauge with $A_+ = A_- = 0$, the only nonzero field strengths at the shock front are $F_{ui} = \partial_u A_i$. Since $g_{uu} = 0$ in double-null coordinates, the stress tensor component sourcing the shift is

$$T_{uu}(u, \mathbf{x}_\perp) = \sum_{i=1,2} (\partial_u A_i)^2 = \hbar\omega\, f(u)\, g_\sigma(r), \tag{84}$$

where $f(u)$ is a normalised longitudinal profile ($\int_{-\infty}^{\infty} f\,du = 1$) encoding the photon wavepacket, and

$$g_\sigma(r) = \frac{1}{\pi\sigma^2}\, e^{-r^2/\sigma^2}, \qquad \sigma \sim \frac{c}{\omega}, \tag{85}$$

is a Gaussian transverse profile normalised so that $\int g_\sigma\, d^2x_\perp = 1$. The total null energy follows immediately from energy conservation:

$$\int T_{uu}\, du\, d^2x_\perp = \hbar\omega \int f\, du \int g_\sigma\, d^2x_\perp = \hbar\omega, \tag{86}$$

independently of the profile shape.

Area increment. Substituting equation (86) into the Raychaudhuri result (46):

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp = 8\pi G\hbar\omega. \tag{87}$$

The area increment is finite, independent of σ, and determined entirely by the photon energy $\hbar\omega$. The regularisation does not alter the area result; it only regularises the spatial distribution of the shift.

The regulated longitudinal shift. Applying the Einstein equations to the perturbed pp-wave metric $ds^2 = -2\,du\,dv + h(\mathbf{x}_\perp)\delta(u)\,du^2 + d^2\mathbf{x}_\perp$ at the shock front, the Ricci component $R_{uu} = 8\pi G T_{uu}$ gives the two-dimensional Poisson equation for the shift $\Delta v = h$:

$$\nabla_\perp^2\, \Delta v = -16\pi G\hbar\omega\, g_\sigma(r) = -\frac{16 G\hbar\omega}{\sigma^2}\, e^{-r^2/\sigma^2}. \tag{88}$$

The total source integral $\int \nabla_\perp^2\, \Delta v\, d^2x_\perp = -16\pi G\hbar\omega$ matches the strict AS case, with the delta function replaced by a smooth Gaussian source. Applying Gauss's theorem in two dimensions to the radially symmetric source:

$$\frac{d\,\Delta v}{dr} = -\frac{8G\hbar\omega}{r}\left(1 - e^{-r^2/\sigma^2}\right), \tag{89}$$

which vanishes at $r = 0$ (the shift is smooth on axis) and approaches $-8G\hbar\omega/r$ for $r \gg \sigma$, recovering the AS form.

Integrating equation (89) and using the identity $\int_0^x (1 - e^{-t})/t\, dt = \gamma + \ln x + E_1(x)$, where γ is the Euler–Mascheroni constant and $E_1(x) = \int_x^\infty e^{-t}/t\, dt$ is the exponential integral, with reference scale $r_0 \gg \sigma$ defined by $\Delta v(r_0) = 0$:

$$\boxed{\Delta v(r) = -4G\hbar\omega\left[\ln\frac{r^2}{r_0^2} + E_1\!\left(\frac{r^2}{\sigma^2}\right)\right].} \tag{90}$$

Limiting behaviour. *On axis, $r \to 0$.* Using the small-argument expansion $E_1(x) \to -\gamma - \ln x$ as $x \to 0^+$, the bracket in (90) has the finite limit

$$\ln\frac{r^2}{r_0^2} + E_1\!\left(\frac{r^2}{\sigma^2}\right) \xrightarrow{r\to 0} \ln\frac{r^2}{r_0^2} - \gamma - \ln\frac{r^2}{\sigma^2} = \ln\frac{\sigma^2}{r_0^2} - \gamma, \tag{91}$$

giving the finite on-axis value

$$\Delta v(0) = 4G\hbar\omega\left[\ln\frac{r_0^2}{\sigma^2} + \gamma\right]. \tag{92}$$

The logarithmic divergence of the strict AS shift at $r = 0$ is resolved: $\Delta v(0)$ grows as $\ln(r_0/\sigma)$ but is finite for any $\sigma > 0$.

Far from axis, $r \gg \sigma$. For large argument, $E_1(r^2/\sigma^2) \to 0$ exponentially, and (90) reduces to

$$\Delta v(r \gg \sigma) \;\approx\; -4G\hbar\omega \ln\frac{r^2}{r_0^2}, \tag{93}$$

recovering the strict AS logarithmic profile.

Recovery of the strict limit. As $\sigma \to 0$ at fixed $r > 0$, $E_1(r^2/\sigma^2) \to 0$ and equation (90) reduces to the strict form (93) pointwise for all $r > 0$, while $\Delta v(0) \to +\infty$ logarithmically via equation (92), recovering the AS singularity on-axis.

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